EART 22201 Igneous minerals and processes

Symmetry and the seven crystal systems



Lesson 1

margaret.hartley@manchester.ac.uk

Symmetry and the seven crystal systems

1.1 The crystal lattice

1.2 The unit cell



margaret.hartley@manchester.ac.uk

What is a mineral?



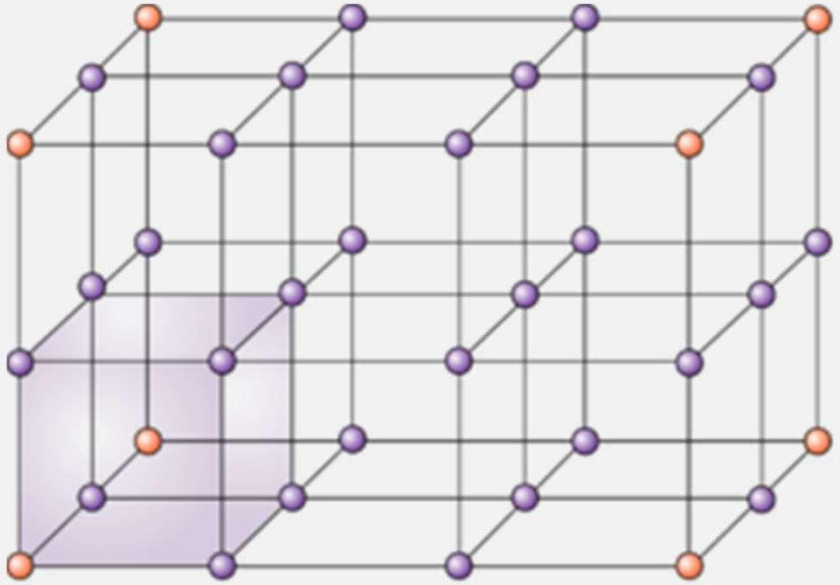
Mineral: A naturally occurring, inorganic solid, with a characteristic chemical composition and a regular internal atomic arrangement which results in a tendency to develop external crystal faces.





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The crystal lattice

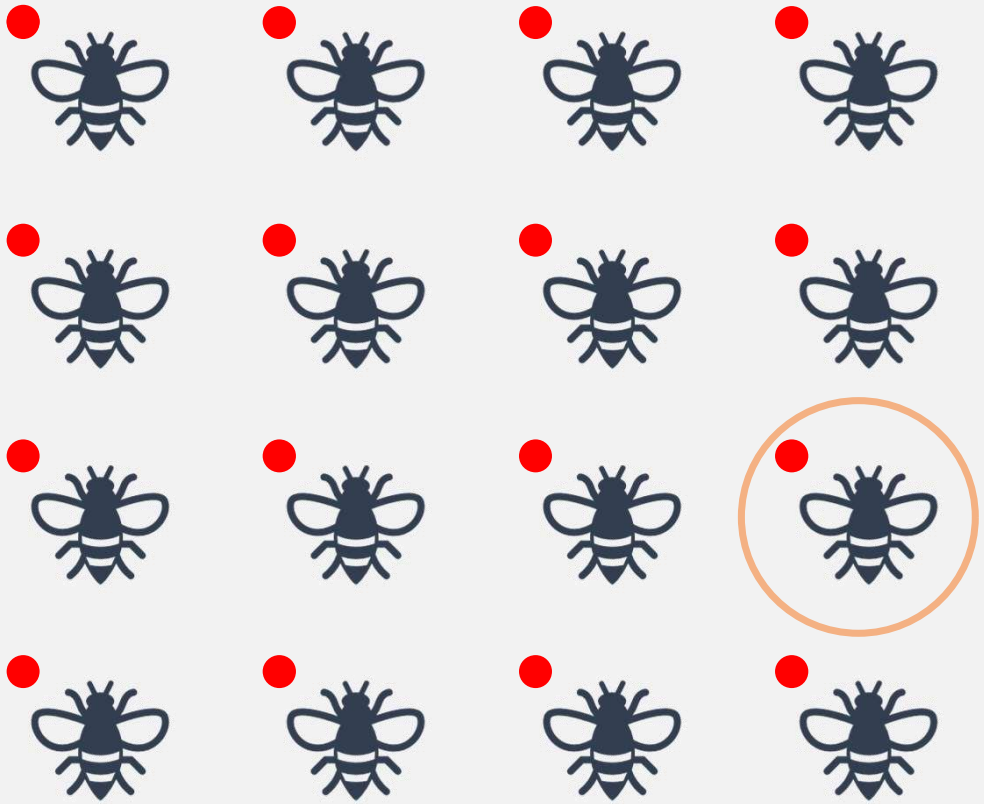


The **crystal lattice** is the 3D arrangement of atoms inside a crystal.

The lattice is an array of hypothetical points in space, each of which has the same physical and chemical environment.

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The crystal lattice



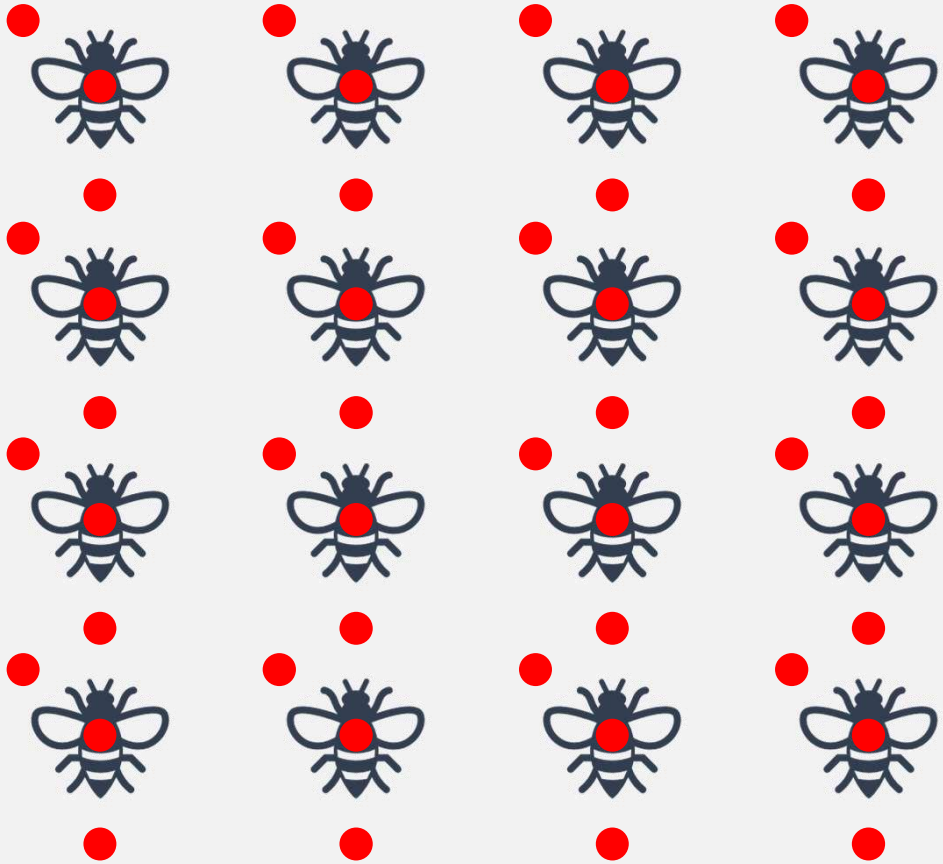
Lattice point: The atom, group of atoms, or pattern unit, that repeats in space.

Motif: The group of atoms that repeats.

Lattice + Motif = **Structure**

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Symmetry and the seven crystal systems

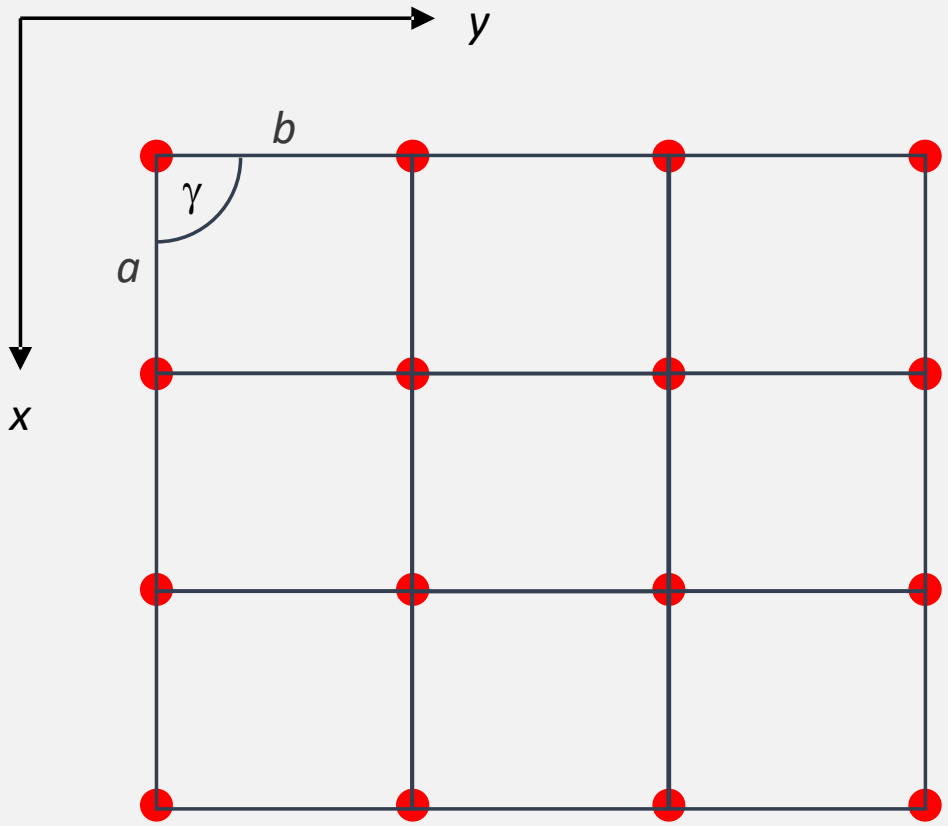
1.1 The crystal lattice

1.2 The unit cell



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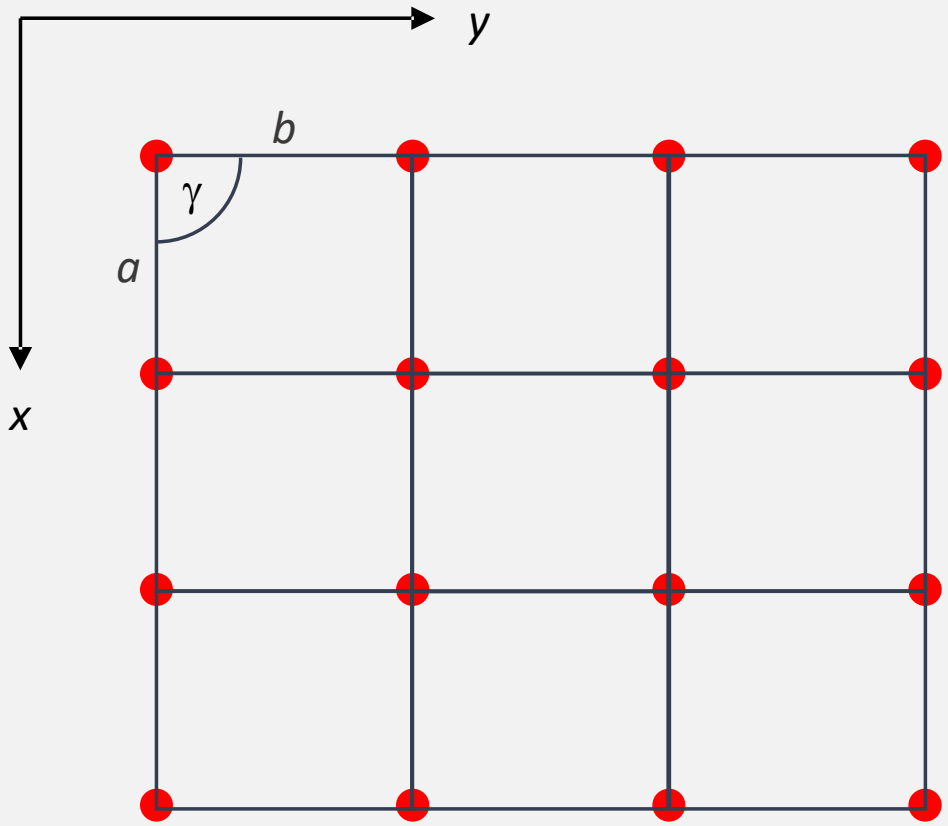
What is a unit cell?

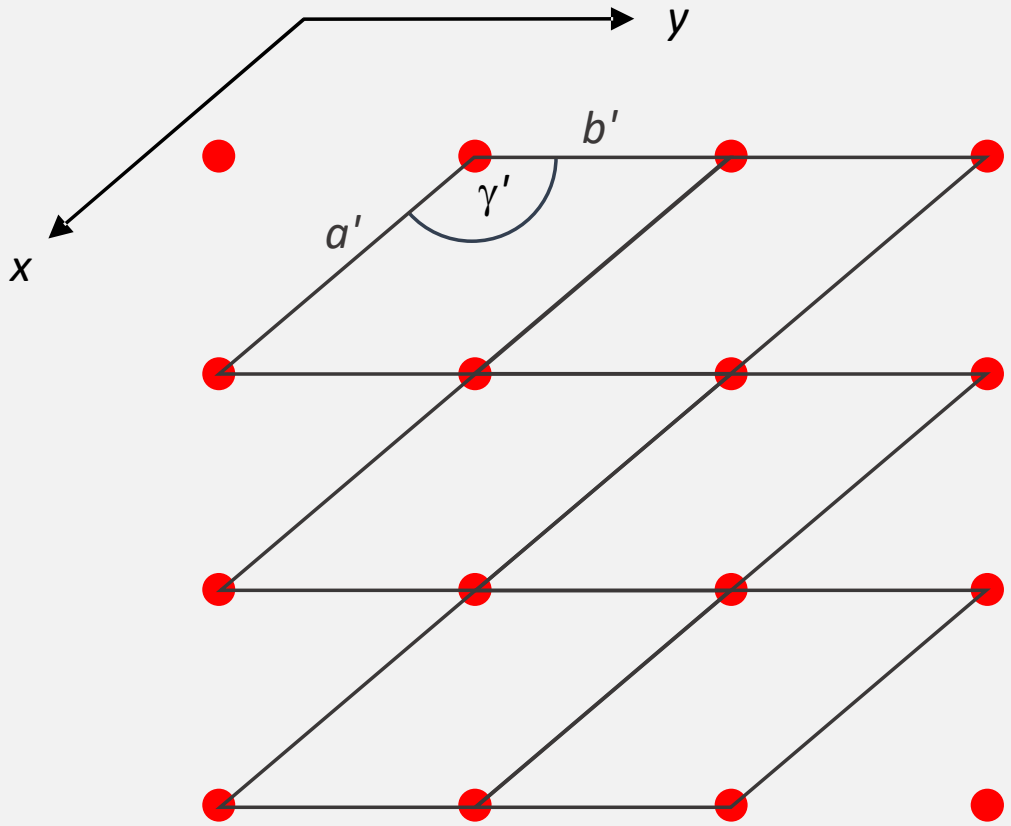


Unit cell: The smallest group of atoms whose geometric arrangement defines the characteristic symmetry of the crystal, and whose repetition in space produces a crystal lattice.

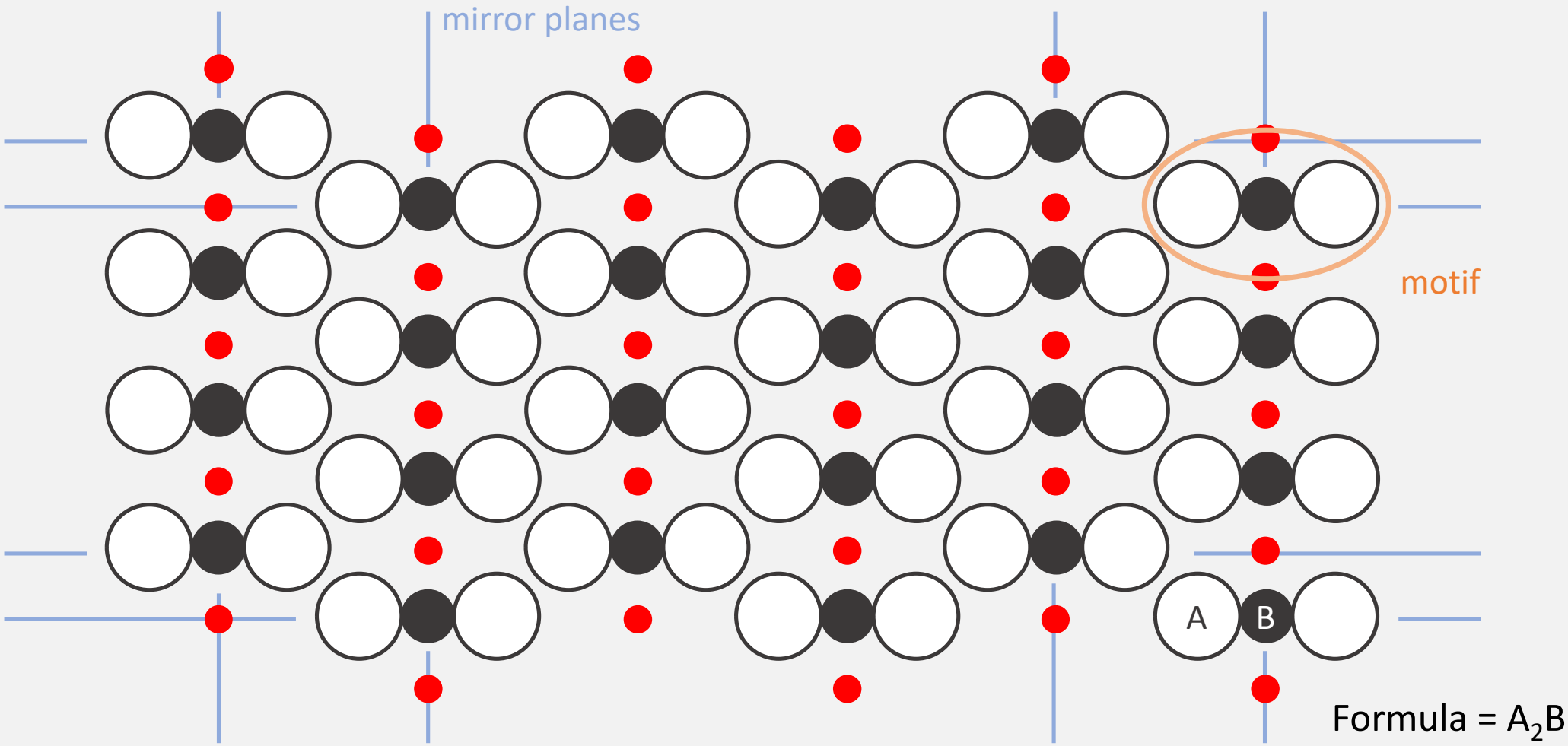
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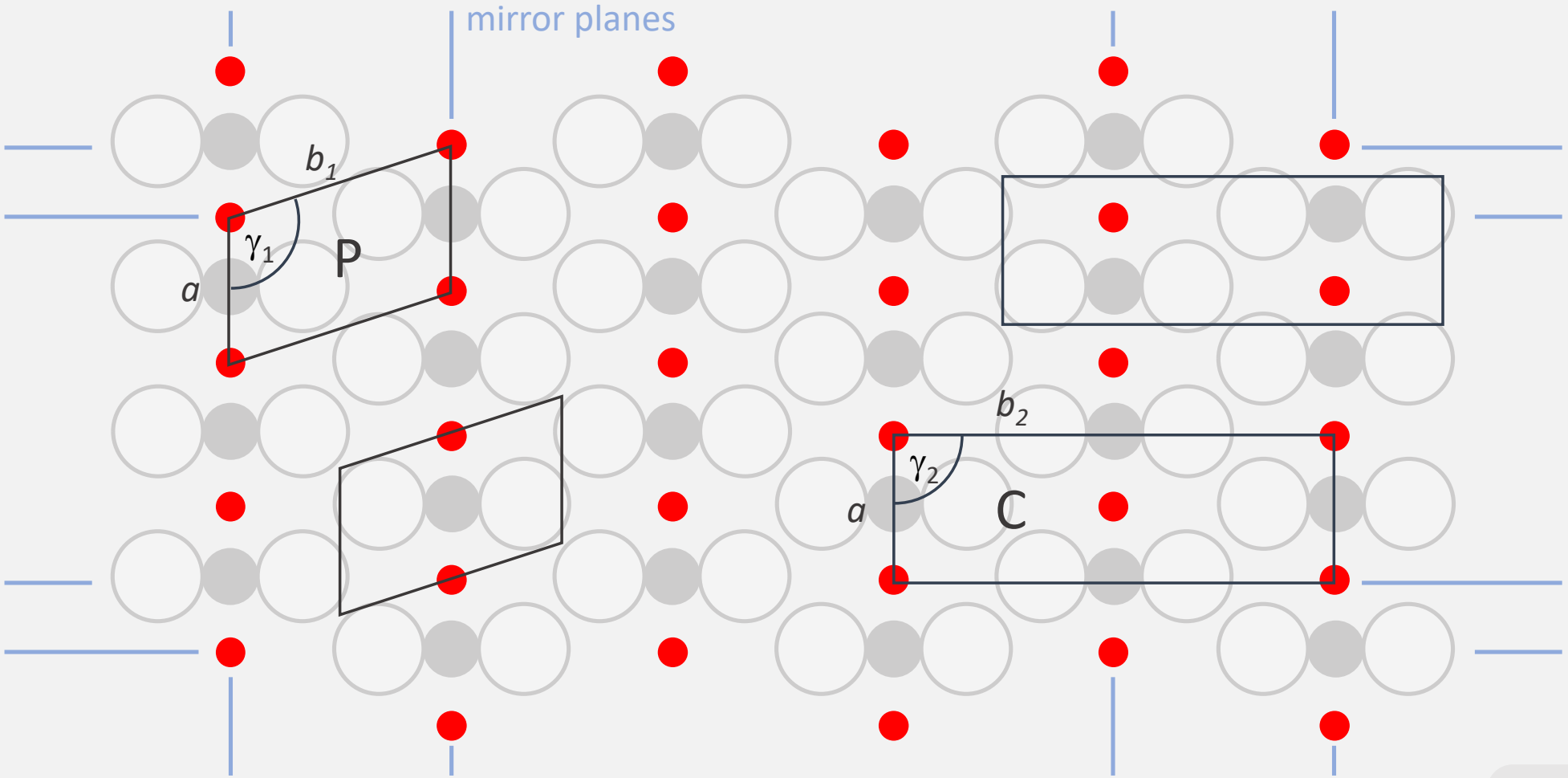


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Formula = A_2B

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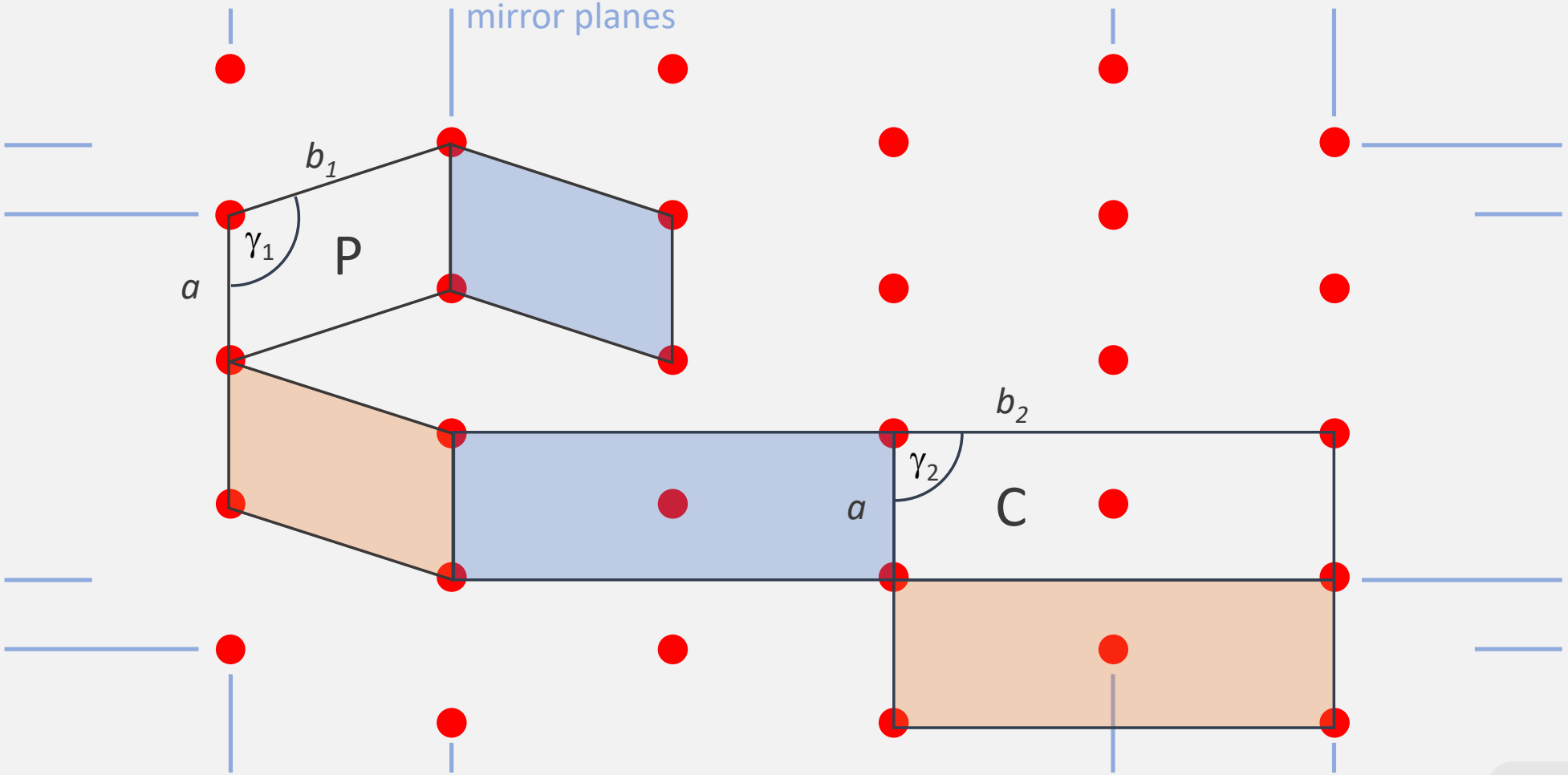


primitive Z=1

centred rectangular Z=2

Z is the total number of formula units contained within the unit cell

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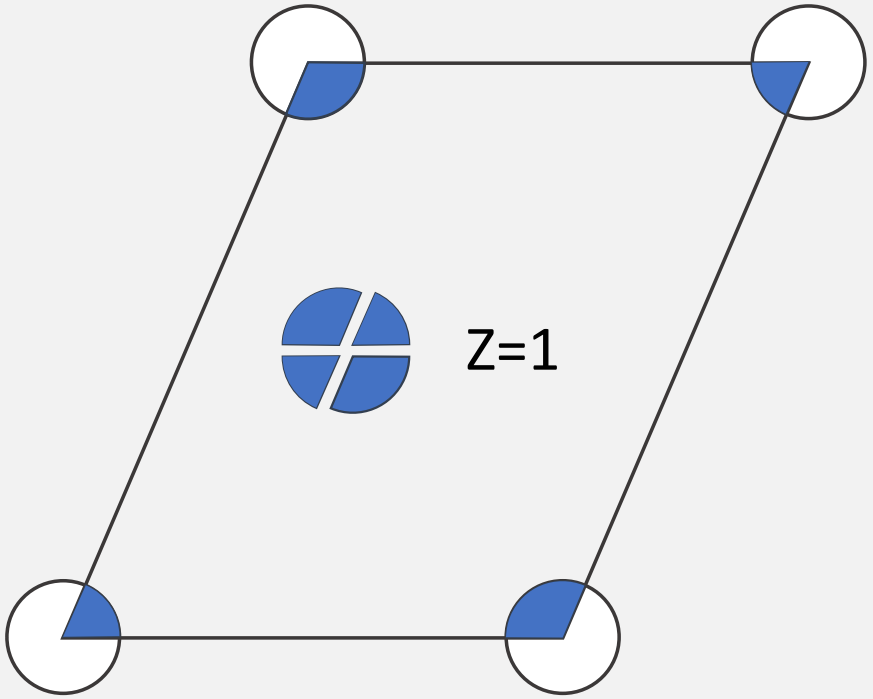
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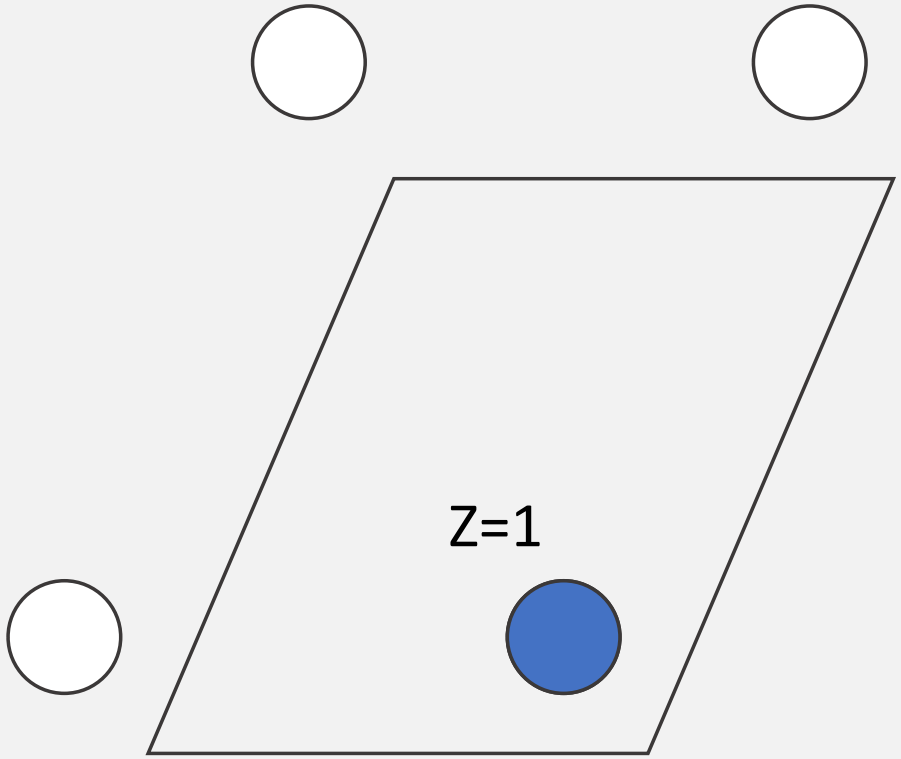
Z is the total number of formula units contained within the unit cell

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How to work out the number of lattice points per 2D unit cell



Method 1: Define the unit cell so that the lattice points sit at the corners. Add up the fractions of each lattice point contained inside the unit cell.



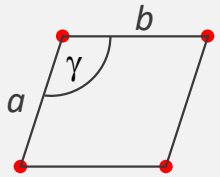
Method 2: Translate the unit cell so that the lattice points are contained within the unit cell. Count up the total number of enclosed lattice points.

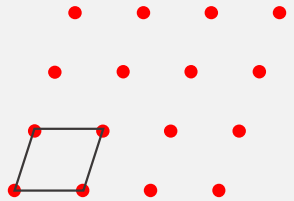
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Types of 2D unit cell

oblique

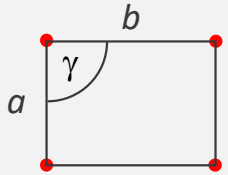
$a \neq b, \gamma \neq 90°$

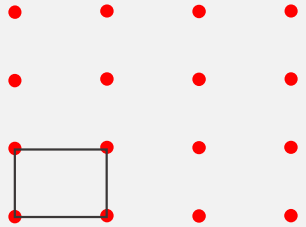




rectangular

$a \neq b, \gamma = 90°$

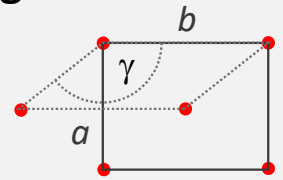


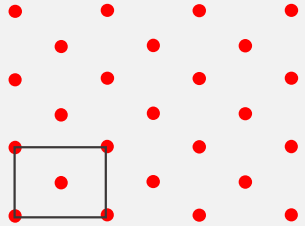


centred rectangular

$a \neq b, \gamma = 90°$

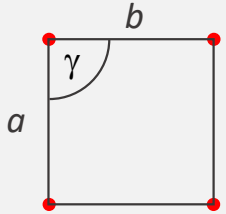
$|\cos \gamma| = b/2a$

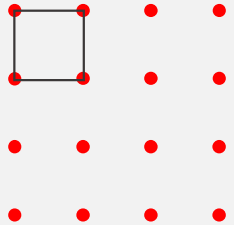




square

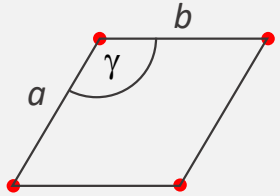
$a = b, \gamma = 90°$

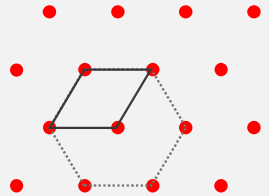




rhombohedral

$a = b, \gamma = 120°$





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Symmetry and the seven crystal systems

You should now be able to:

- Explain the concept of the unit cell
- Recognise different types of unit cell
- Select the most appropriate unit cell for a given 2D lattice



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Symmetry and the seven crystal systems

2.1 Lattice parameters in 3D

2.2 Seven crystal systems

2.3 Unit cells in 3D

2.4 Bravais lattices



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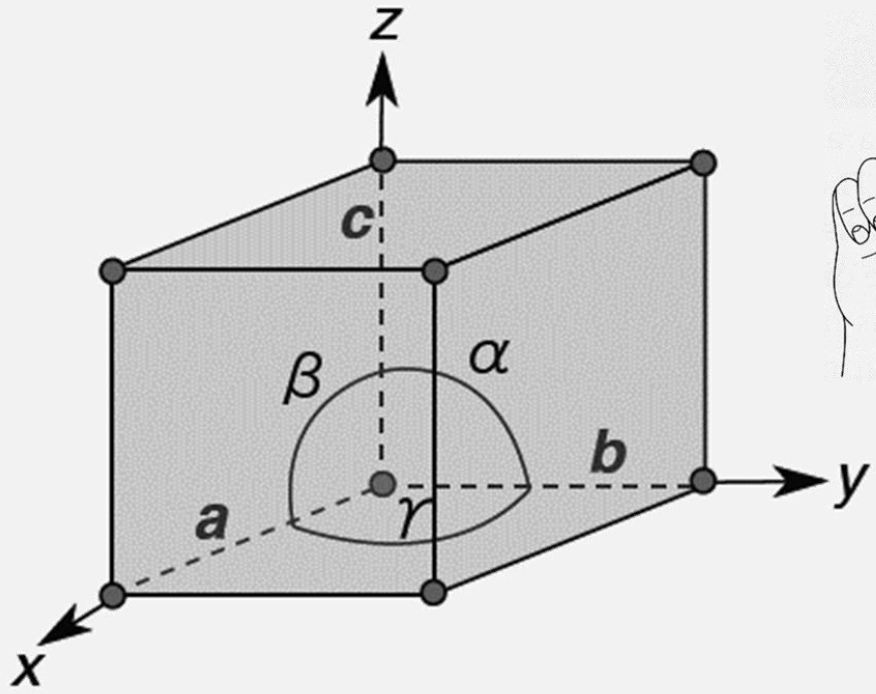
Lattices in 3D: Lattice parameters

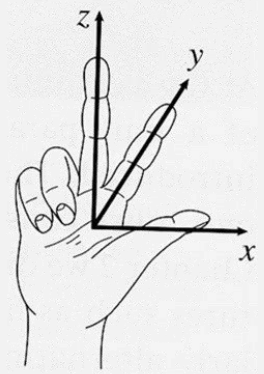
Lattice parameters are the physical dimensions of the unit cell in the crystal lattice.

To fully describe a 3D lattice, we need to know:

- The lengths of the lattice edges: *a*, *b*, and *c*
- The angles between the lattice sides: α, β, and γ

a ∥ *x*	α = *y*^*z*
b ∥ *y*	β = *x*^*z*
c ∥ *z*	γ = *x*^*y*





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Symmetry and the seven crystal systems

2.1 Lattice parameters in 3D

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2.3 Unit cells in 3D

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Crystal system	Lattice parameters	
Cubic	$a = b = c$	$\alpha = \beta = \gamma = 90°$
Trigonal (rhombohedral)	$a = b = c$	$\alpha = \beta = \gamma = 120°$
Tetragonal	$a = b \neq c$	$\alpha = \beta = \gamma = 90°$
Hexagonal	$a = b \neq c$	$\alpha = \beta = 90°$, $\gamma = 120°$
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Monoclinic	$a \neq b \neq c$	$\alpha = \gamma = 90°$, $\beta > 90°$
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most symmetry



least symmetry

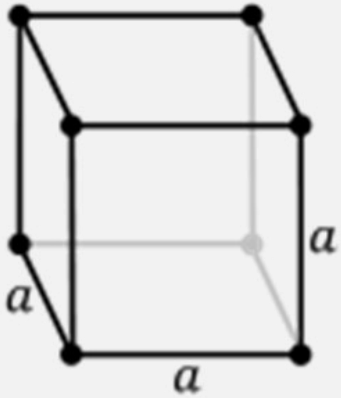
Seven crystal systems

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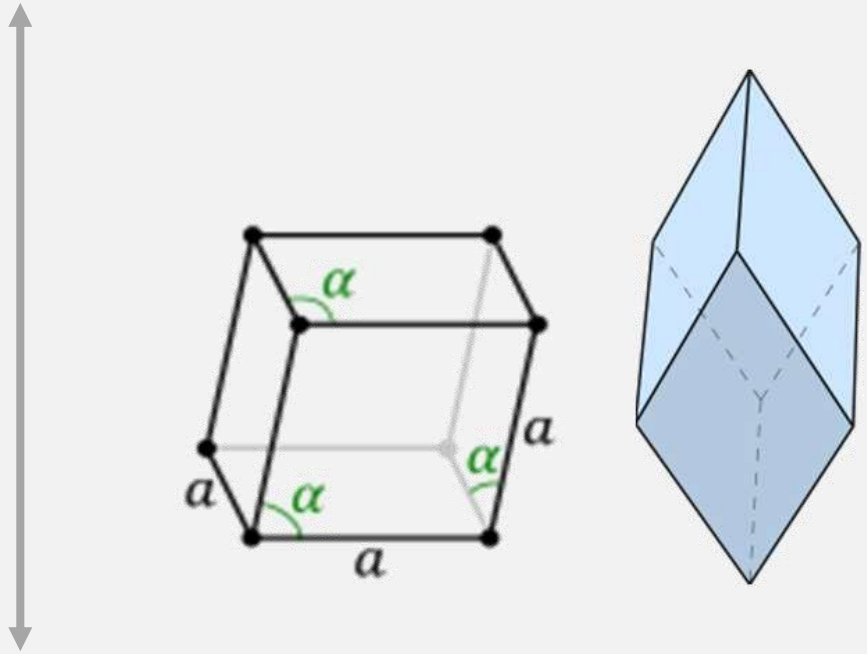
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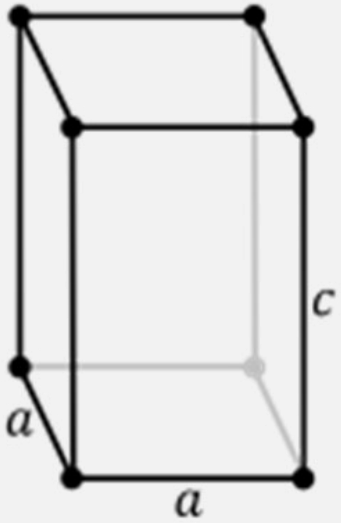
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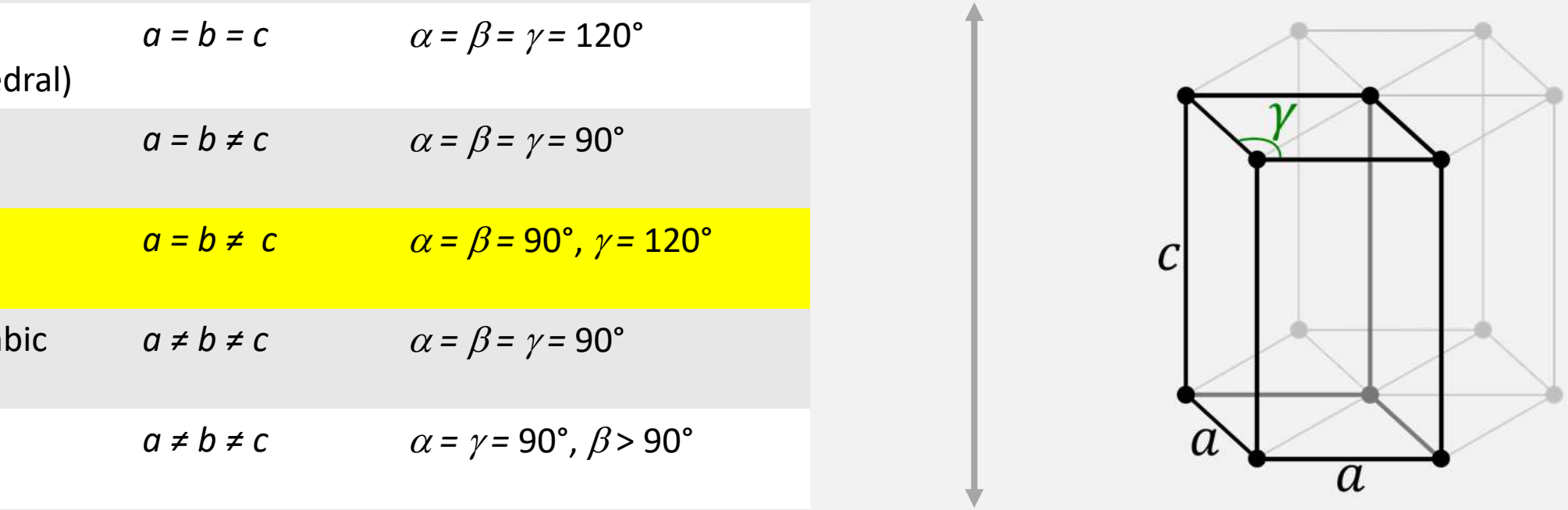
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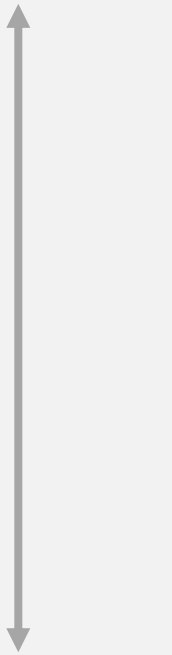


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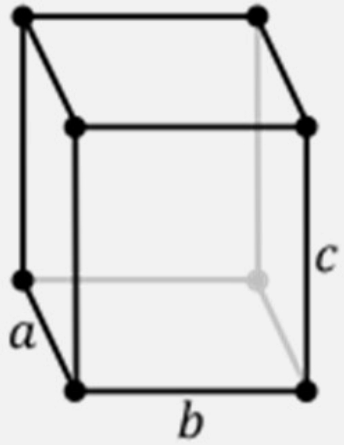
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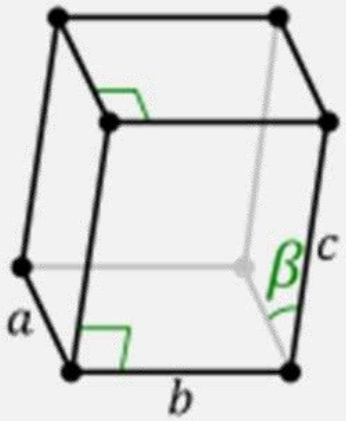


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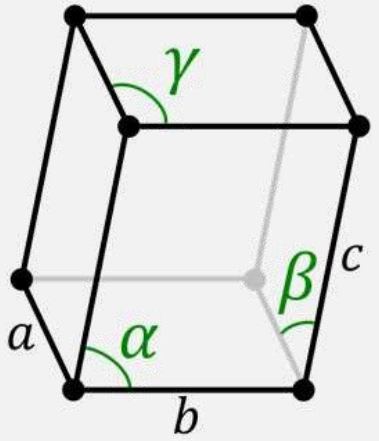


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Symmetry and the seven crystal systems

2.1 Lattice parameters in 3D

2.2 Seven crystal systems

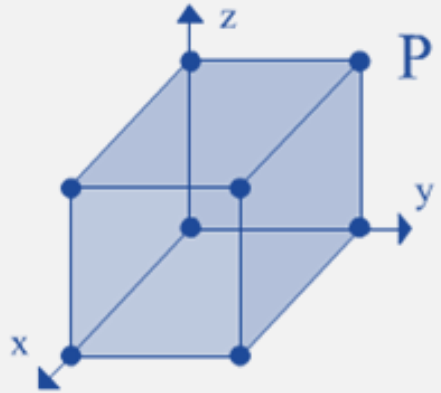
2.3 Unit cells in 3D

2.4 Bravais lattices

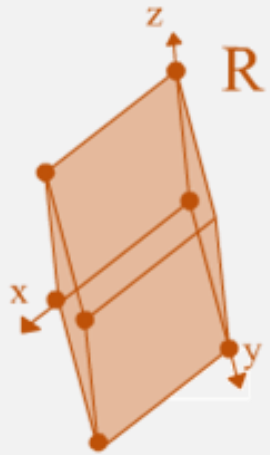


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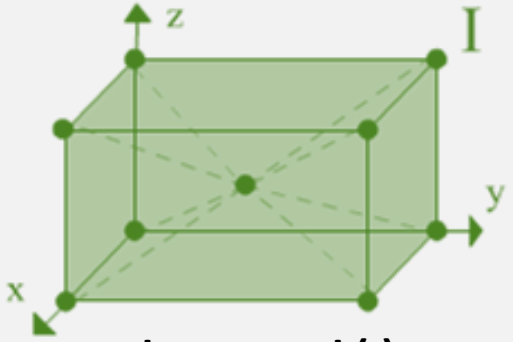
The five types of 3D unit cell



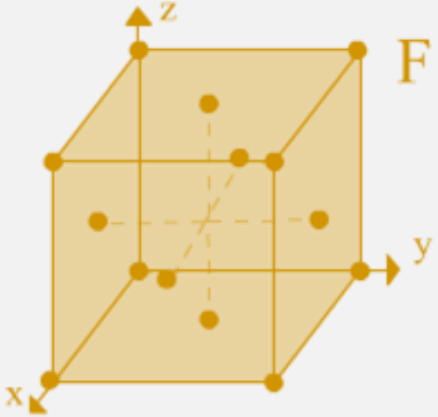
Primitive (P)
Lattice points only at corners



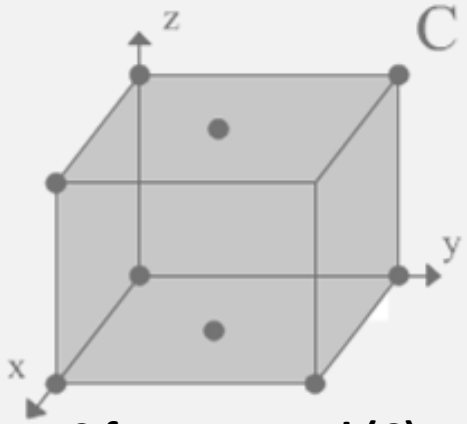
Rhombohedral (R)
Lattice points only at corners



Body-centred (I)
Lattice points at corners and at the intersection of the body diagonals



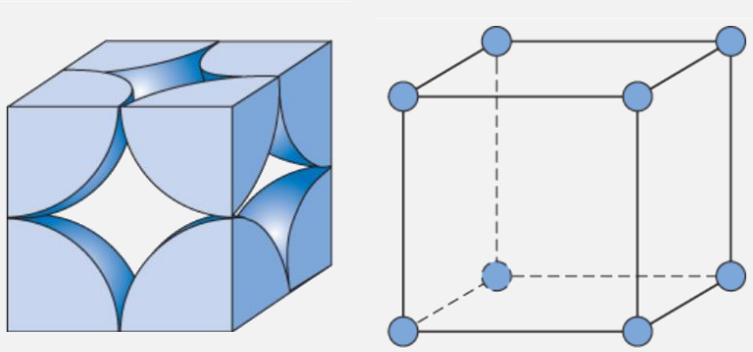
Face-centred (F)
Lattice points at corners and at the centre of every face



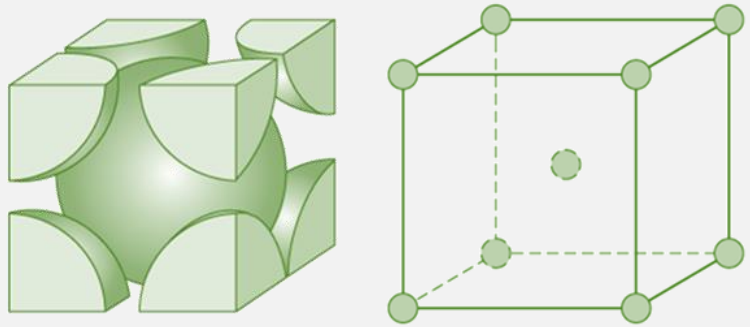
C face-centred (C)
Lattice points at corners, and at the centre of one pair of opposite faces

margaret.hartley@manchester.ac.uk

How to work out the number of lattice points per 3D unit cell



Primitive (P) and **Rhombohedral (R)**
Corners: 8 × ⅛
1 lattice point per unit cell

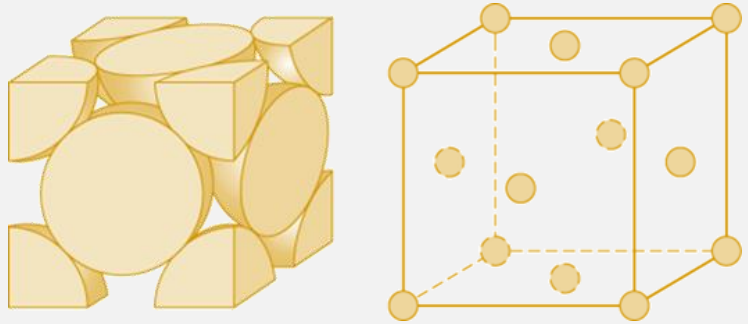


Body-centred (I)
Corners: 8 × ⅛
Fully enclosed: 1
2 lattice points per unit cell

Lattice points on corners = ⅛

Lattice points on faces = ½

Lattice points fully enclosed = 1



Face-centred (F)
Corners: 8 × ⅛
Faces: 6 × ½
4 lattice points per unit cell

See if you can work out the number of lattice points in a C face-centred unit cell

margaret.hartley@manchester.ac.uk

Symmetry and the seven crystal systems

2.1 Lattice parameters in 3D

2.2 Seven crystal systems

2.3 Unit cells in 3D

2.4 Bravais lattices



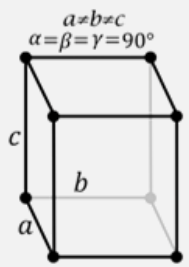
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Crystal system	Lattice parameters		Possible unit cell types
Cubic	$a = b = c$	$\alpha = \beta = \gamma = 90°$	P, I, F
Trigonal (rhombohedral)	$a = b = c$	$\alpha = \beta = \gamma \neq 90°$	R
Tetragonal	$a = b \neq c$	$\alpha = \beta = \gamma = 90°$	P, I
Hexagonal	$a = b \neq c$	$\alpha = \beta = 90°$, $\gamma = 120°$	P
Orthorhombic	$a \neq b \neq c$	$\alpha = \beta = \gamma = 90°$	P, C, I, F
Monoclinic	$a \neq b \neq c$	$\alpha = \gamma = 90°$, $\beta > 90°$	P, C
Triclinic	$a \neq b \neq c$	$\alpha \neq \beta \neq \gamma \neq 90°$	P

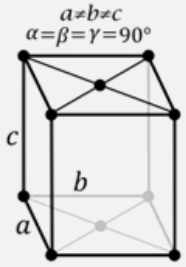
The seven crystal systems and five lattice types can be combined in 14 different ways. These are called **Bravais lattices**.

margaret.hartley@manchester.ac.uk

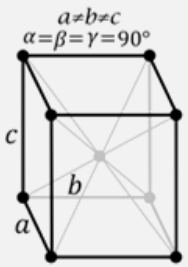
14 Bravais lattices



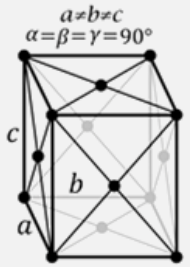
Primitive



C face-centred

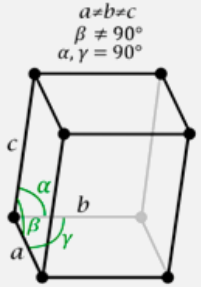


Body-centred

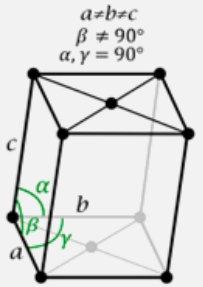


Face-centred

Orthorhombic

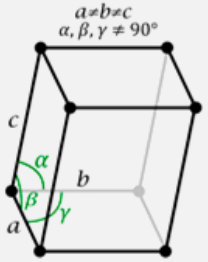


Primitive



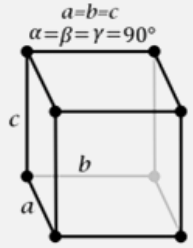
C face-centred

Monoclinic

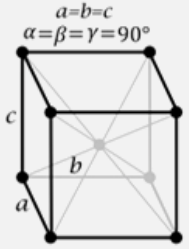


Primitive

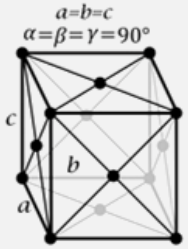
Triclinic



Primitive

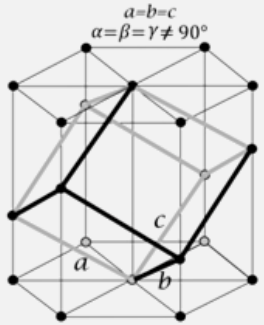


Body-centred



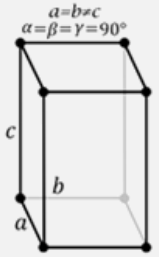
Face-centred

Cubic

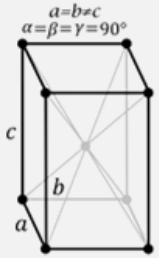


Rhombohedral

Trigonal

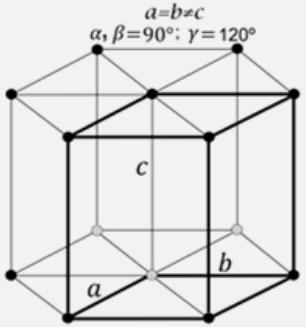


Primitive



Body-centred

Tetragonal



Primitive

Hexagonal

margaret.hartley@manchester.ac.uk

Symmetry and the seven crystal systems

You should now be able to:

- List the seven crystal systems and their different lattice parameters
- Recognise the five types of 3D unit cell
- Explain the concept of Bravais lattices



margaret.hartley@manchester.ac.uk

Symmetry and the seven crystal systems

3.1 Symmetry operators: Reflection, rotation, inversion

3.2 Minimum symmetry requirements in the seven crystal systems

3.3 Combining symmetry operators

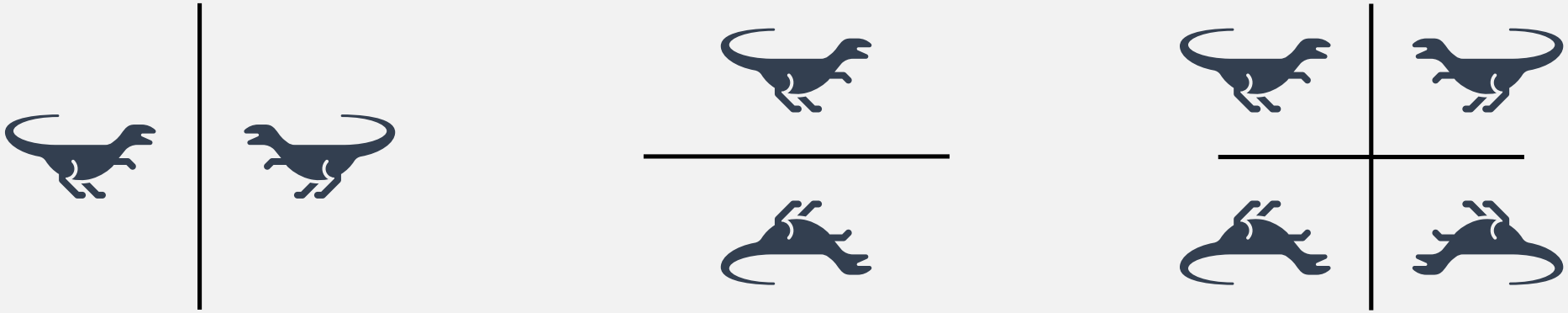
3.4 Crystal form and habit



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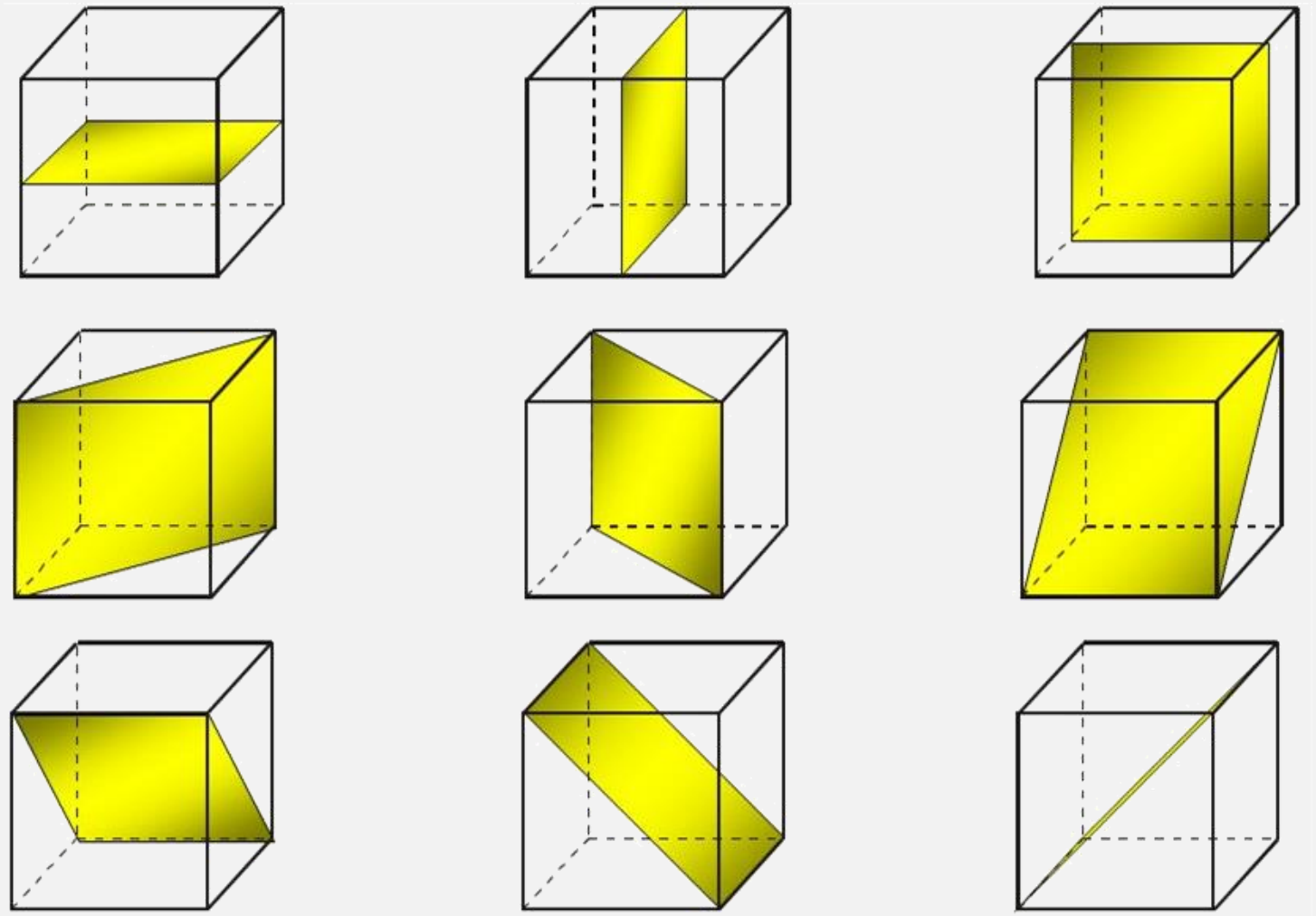
Reflection

mirror plane



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Example: reflection in a cube



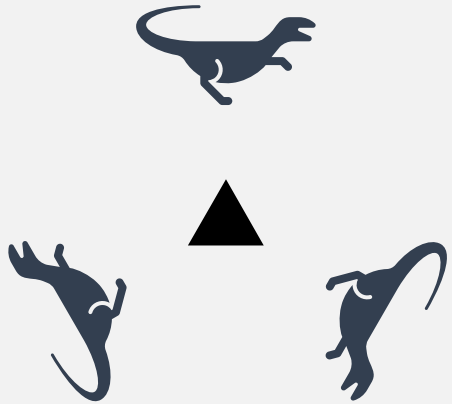
9 mirror planes in a cube

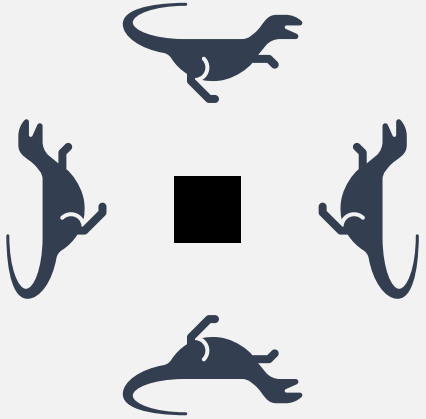
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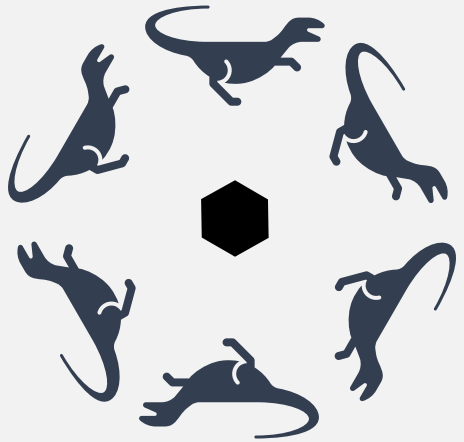
Rotation

Rotation order	Name	Symbol	Rotation angle
1	identity	*(none)*	360
2	diad		180
3	triad		120
4	tetrad		90
6	hexad		60





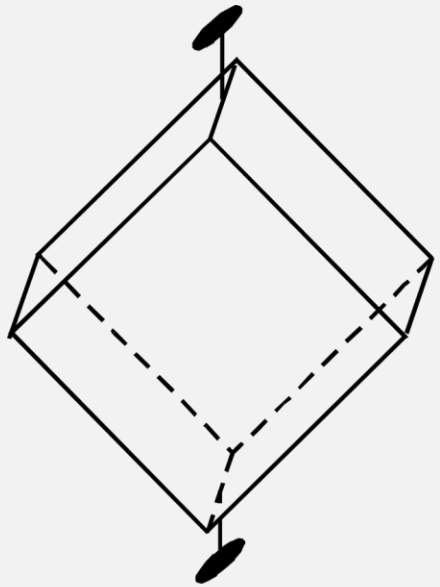




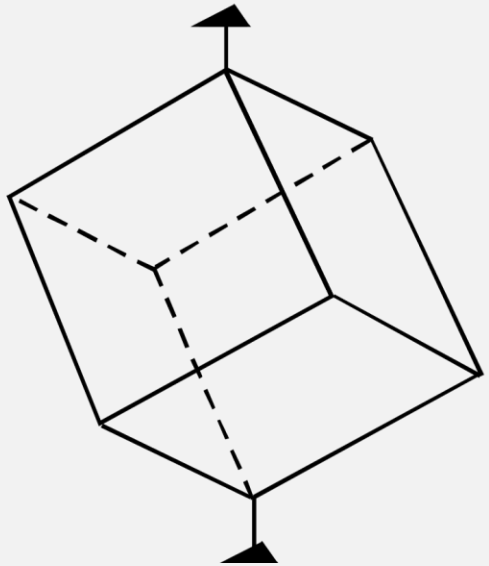
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Example: rotational symmetry in a cube

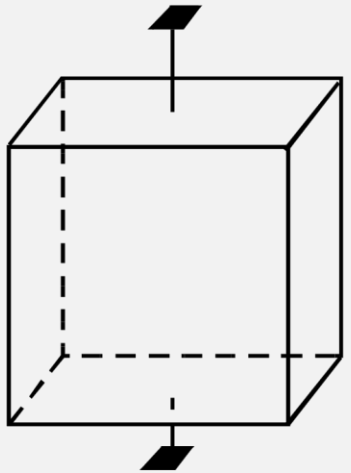
diad



triad



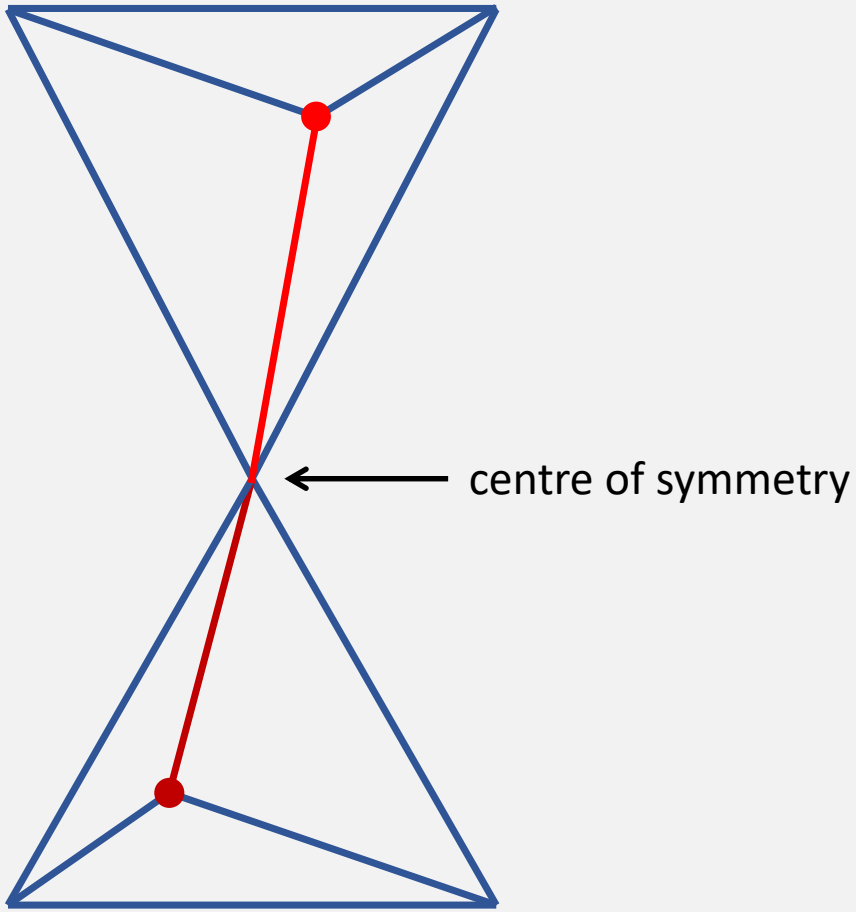
tetrad

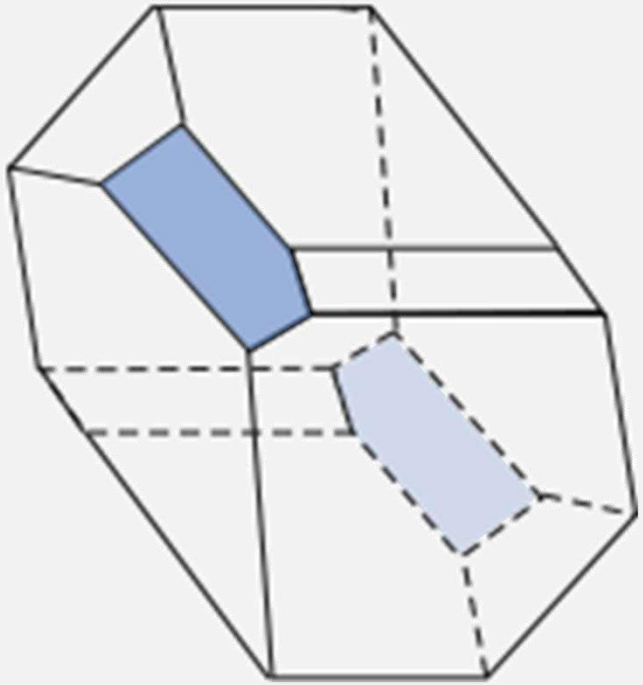


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Centre of symmetry

Also called centre of inversion



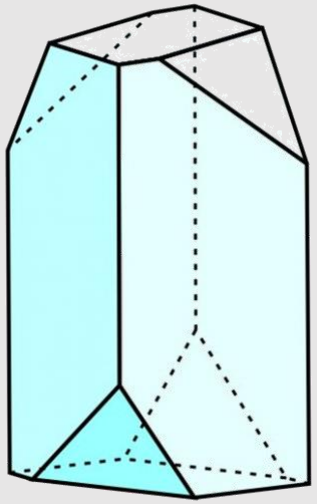


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Rotoinversion

Rotation order X	Name	Symbol	Rotoinversion
1	identity	*(none)*	
2	diad		
3	triad		
4	tetrad		
6	hexad		

A centre of symmetry can be combined with rotation to create an additional class of symmetry called rotoinversion.



This object has 4-fold rotoinversion: the corners and faces are related by rotation about 90° and inverting through the centre of symmetry.

margaret.hartley@manchester.ac.uk

Symmetry and the seven crystal systems

3.1 Symmetry operators: Reflection, rotation, inversion

3.2 Minimum symmetry requirements in the seven crystal systems

3.3 Combining symmetry operators

3.4 Crystal form and habit



margaret.hartley@manchester.ac.uk

Crystal system	Lattice parameters		Essential symmetry	
Cubic	$a = b = c$	$\alpha = \beta = \gamma = 90°$	4 triads	
Trigonal (rhombohedral)	$a = b = c$	$\alpha = \beta = \gamma \neq 90°$	1 triad	
Tetragonal	$a = b \neq c$	$\alpha = \beta = \gamma = 90°$	1 tetrad	
Hexagonal	$a = b \neq c$	$\alpha = \beta = 90°$, $\gamma = 120°$	1 hexad	
Orthorhombic	$a \neq b \neq c$	$\alpha = \beta = \gamma = 90°$	3 mutually perpendicular diads	
Monoclinic	$a \neq b \neq c$	$\alpha = \gamma = 90°$, $\beta > 90°$	1 diad	
Triclinic	$a \neq b \neq c$	$\alpha \neq \beta \neq \gamma \neq 90°$	none	

margaret.hartley@manchester.ac.uk

Symmetry and the seven crystal systems

3.1 Symmetry operators: Reflection, rotation, inversion

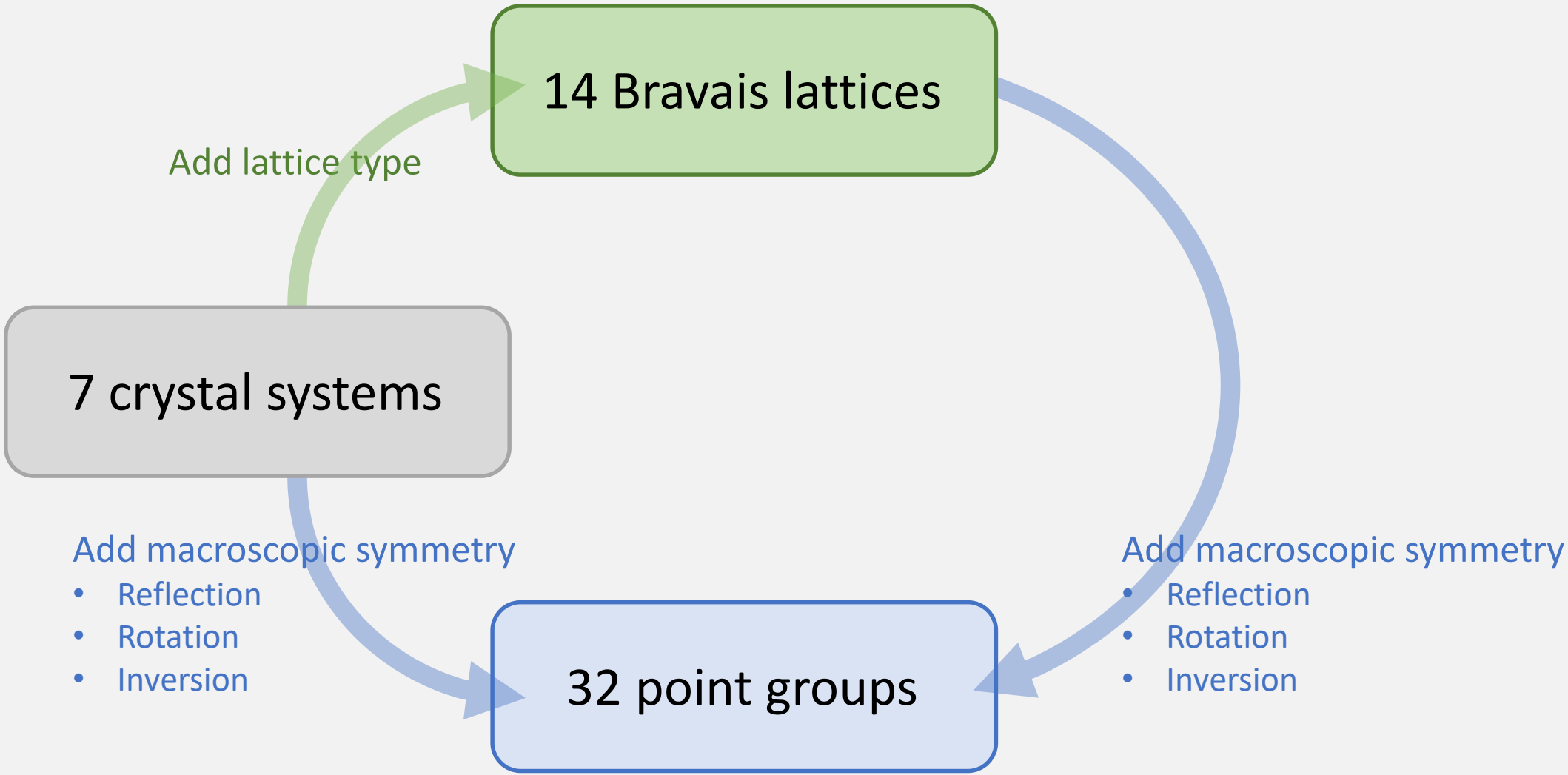
3.2 Minimum symmetry requirements in the seven crystal systems

3.3 Combining symmetry operators

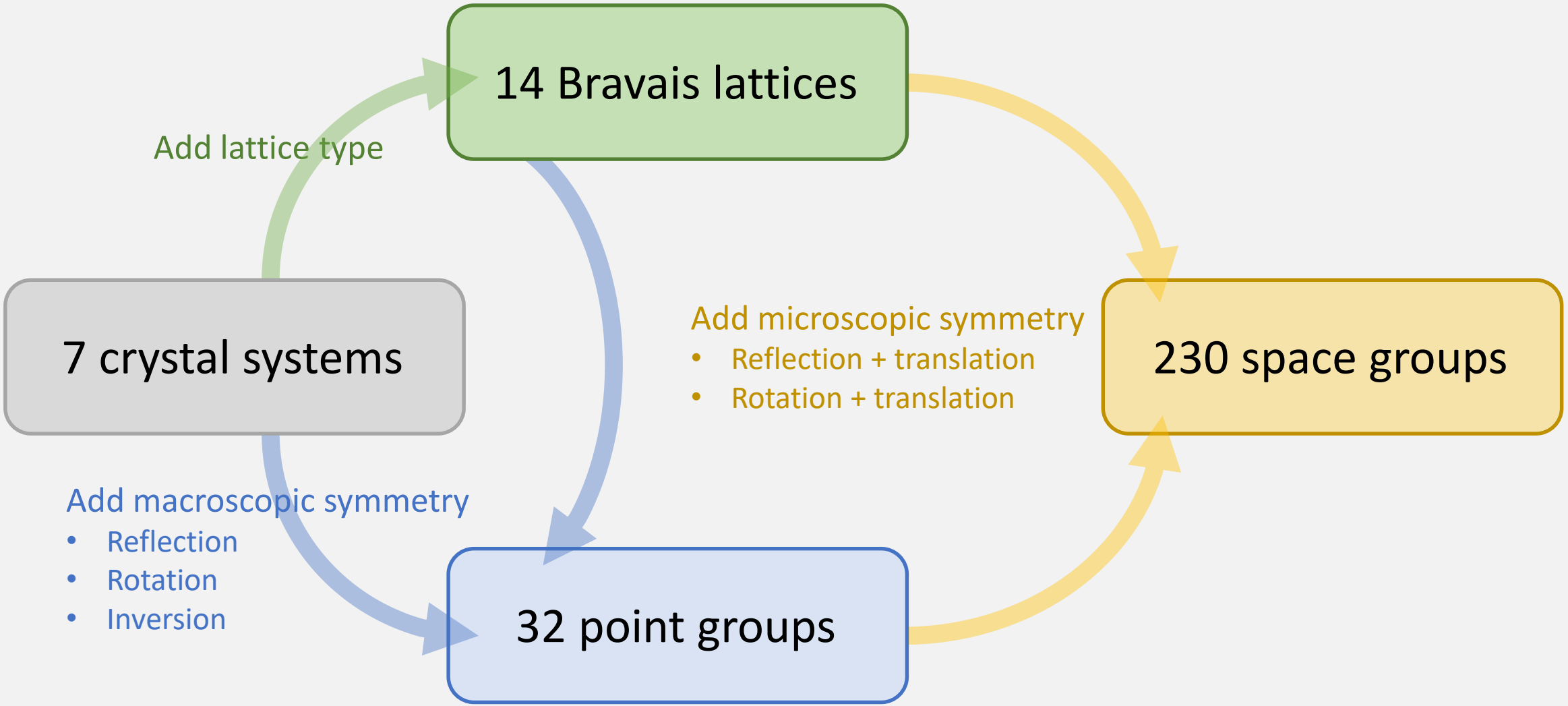
3.4 Crystal form and habit



margaret.hartley@manchester.ac.uk



margaret.hartley@manchester.ac.uk



margaret.hartley@manchester.ac.uk

Symmetry and the seven crystal systems

3.1 Symmetry operators: Reflection, rotation, inversion

3.2 Minimum symmetry requirements

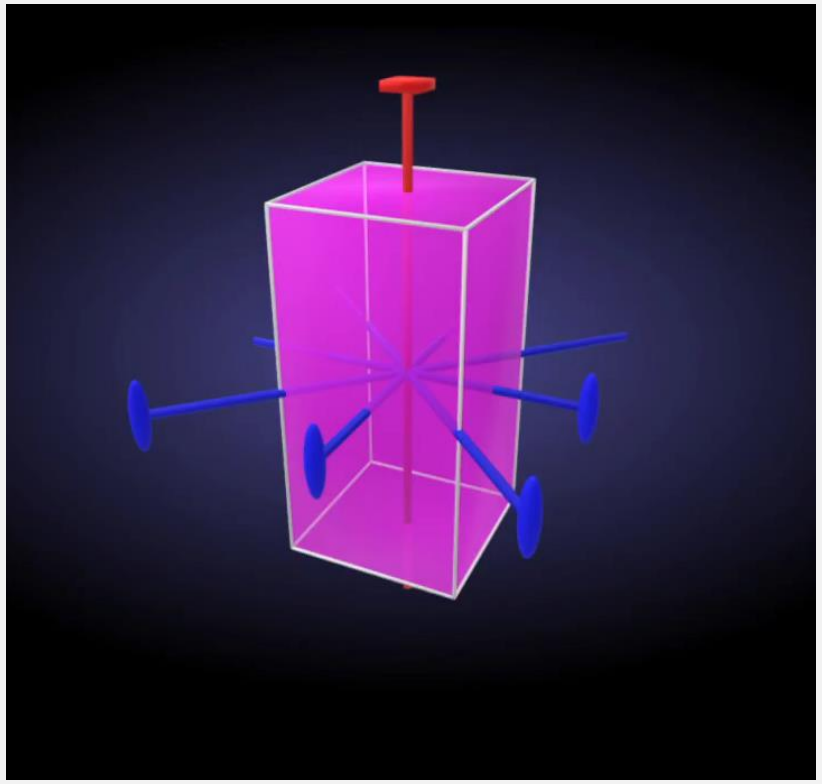
3.3 Combining symmetry operators

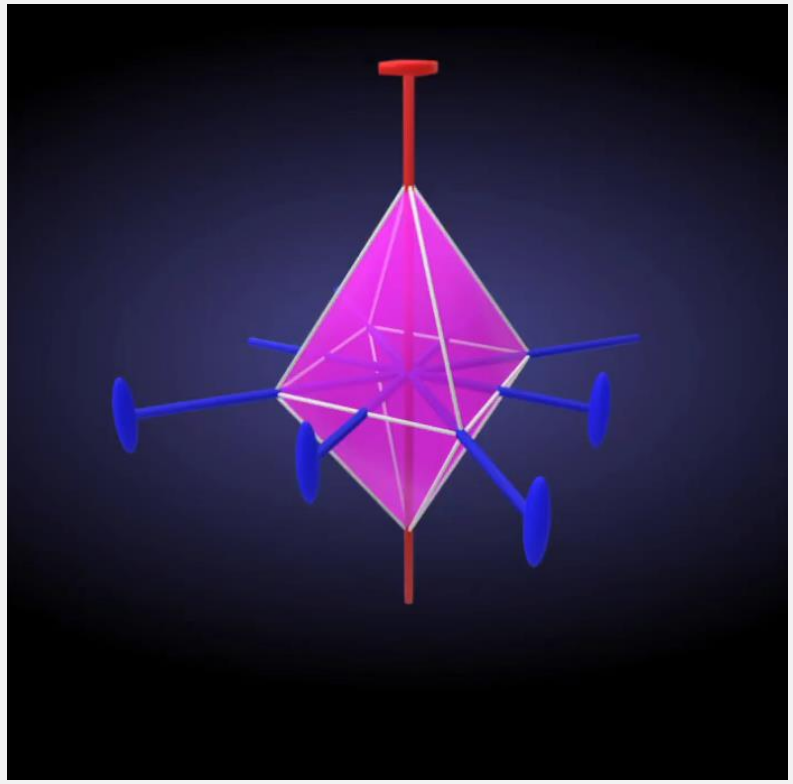
3.4 Crystal form and habit



margaret.hartley@manchester.ac.uk

External form matches the underlying crystal symmetry





Both these crystal forms have:

- One tetrad
- Four diads
- Five mirror planes

margaret.hartley@manchester.ac.uk

External form matches the underlying crystal symmetry



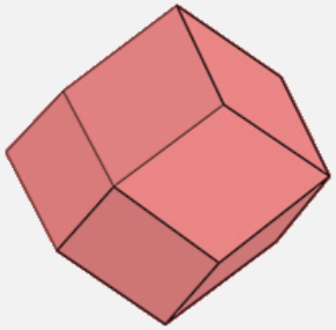
This crystal form has:

- One tetrad
- One mirror plane

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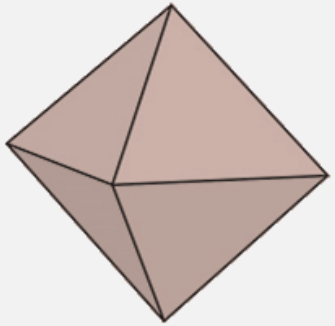
Every mineral has an ideal external form

garnet



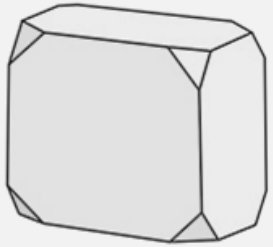
cubic

spinel



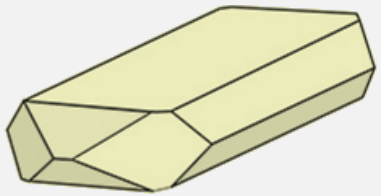
cubic

halite



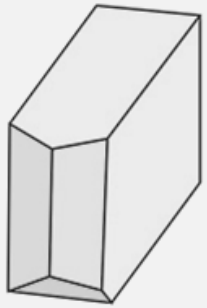
cubic

barite



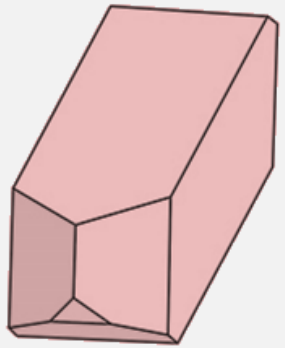
orthorhombic

albite



triclinic

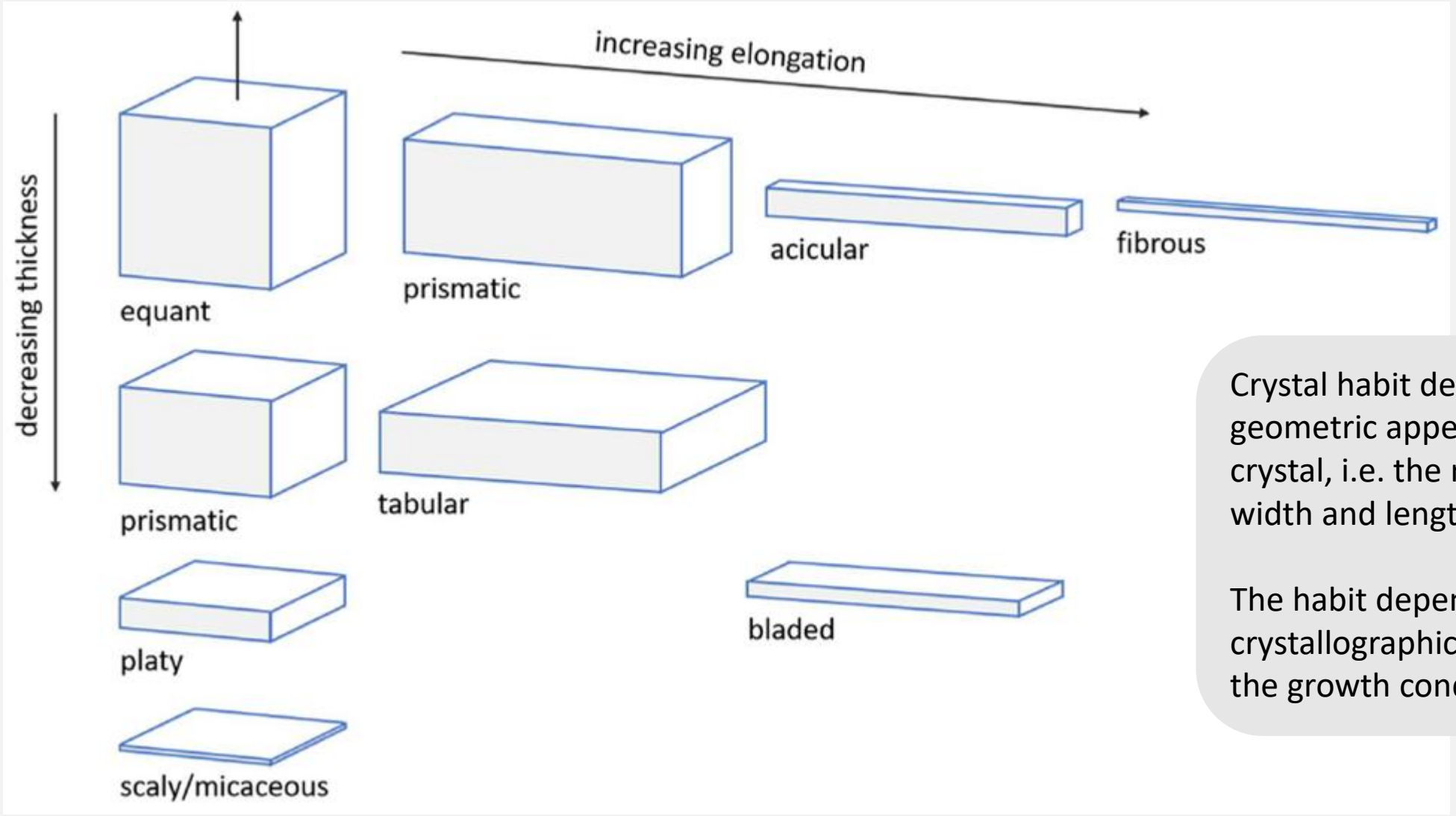
orthoclase



monoclinic

margaret.hartley@manchester.ac.uk

Crystal habit



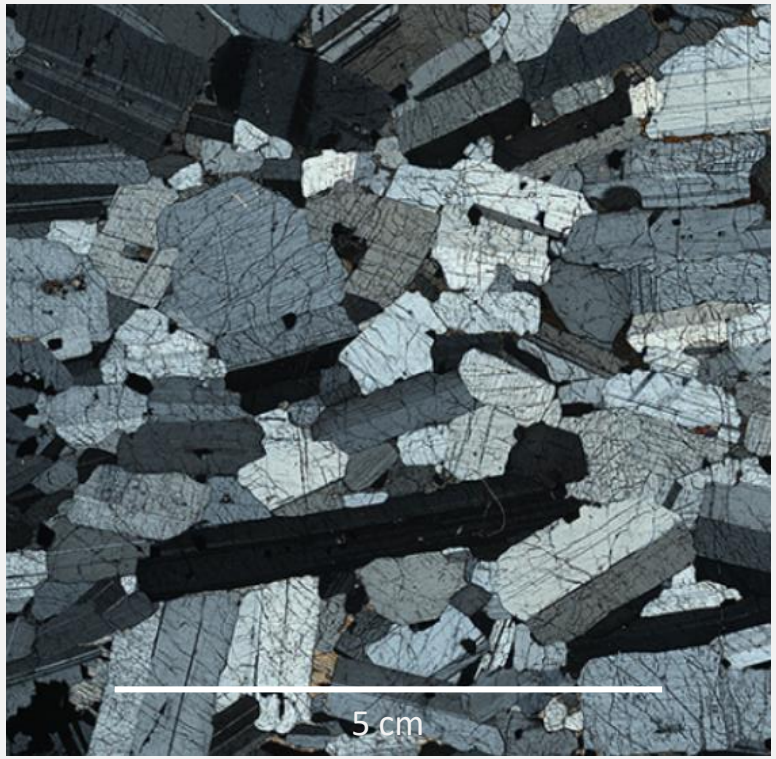
Crystal habit describes the geometric appearance of the crystal, i.e. the ratio of height, width and length.

The habit depends on the crystallographic form and on the growth conditions.

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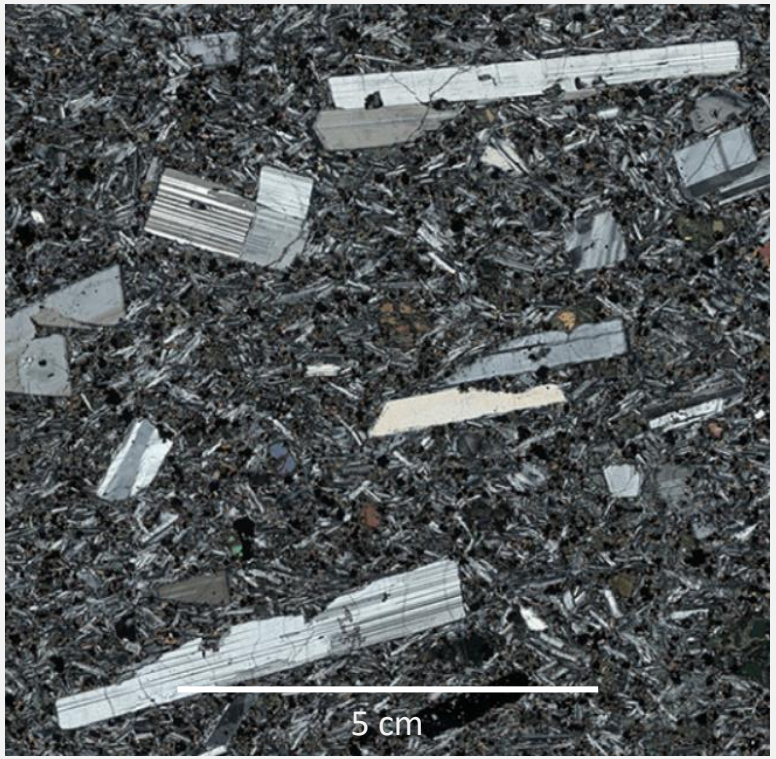
One mineral can show different crystal habits

Plagioclase in a gabbro



Large tabular crystals

Plagioclase in a basalt



Large tabular phenocrysts
Small acicular groundmass crystals

margaret.hartley@manchester.ac.uk

Describing the quality of the crystal shape



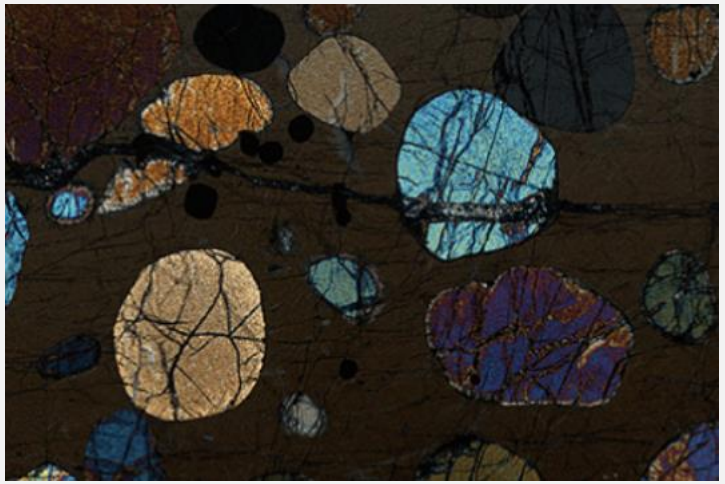
euhedral

subhedral

anhedral



Equant, euhedral olivine in a basalt



Rounded, anhedral olivine in a gabbro

margaret.hartley@manchester.ac.uk

Symmetry and the seven crystal systems

You should now be able to:

- Recognise reflectional, rotational and inversional symmetry
- Explain the concept of a point group
- Describe crystal habit and the quality of crystal form



margaret.hartley@manchester.ac.uk

Symmetry and the seven crystal systems

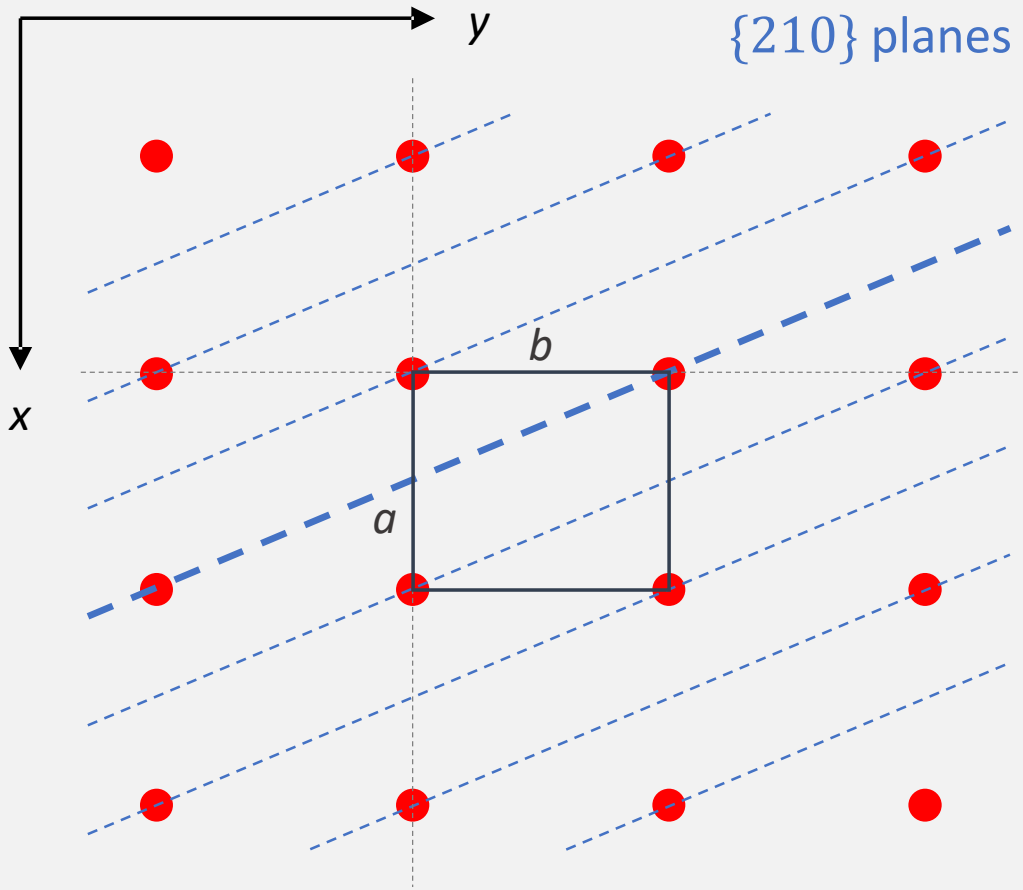
4.1 Miller indices

4.2 Cleavage and twinning in minerals



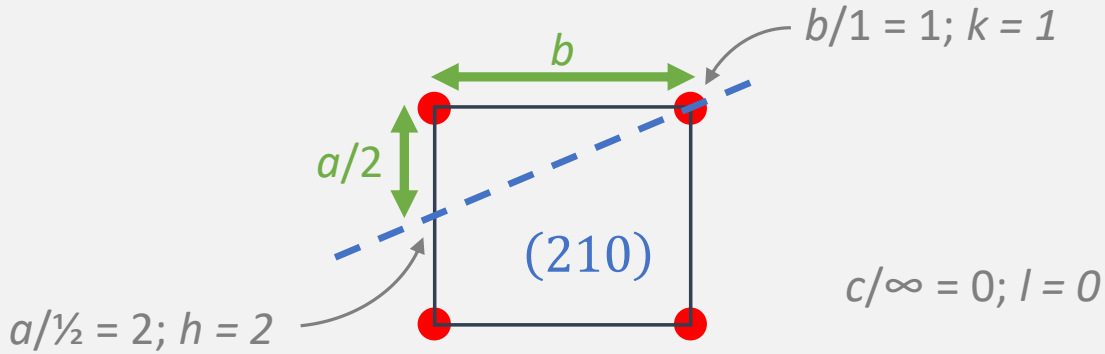
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Miller indices



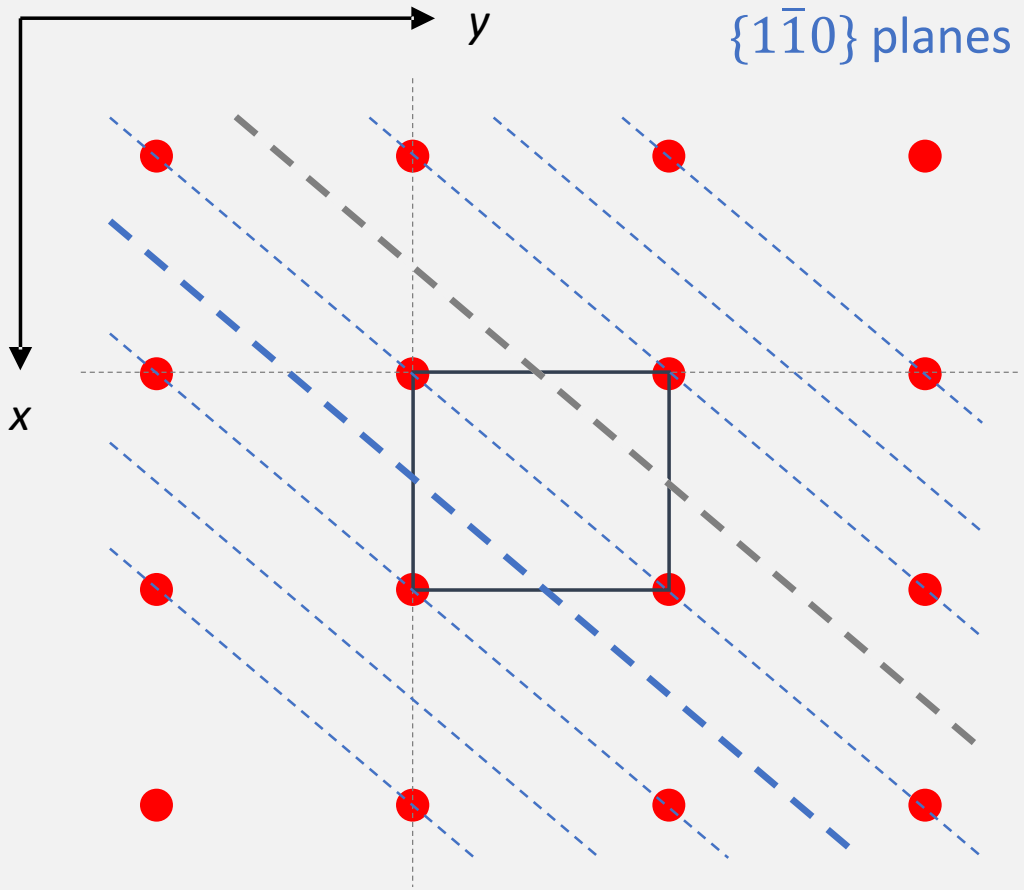
Miller index: *(hkl)*
where *h, k* and *l* are obtained by taking the reciprocal of the axis intercept.

The first lattice plane away from the origin intercepts the x, y, and z axes at distances of *a/h*, *b/k* and *c/l*.



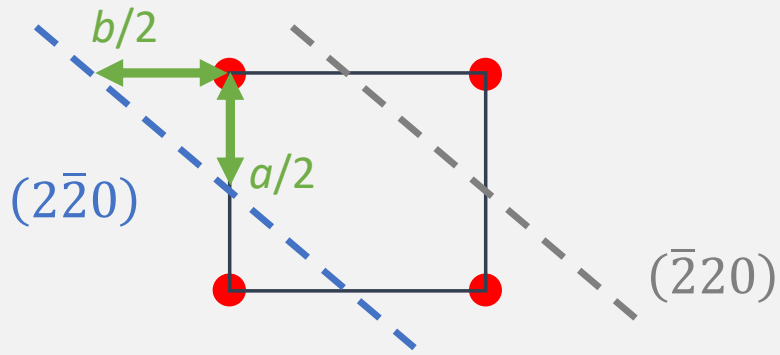
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Miller indices



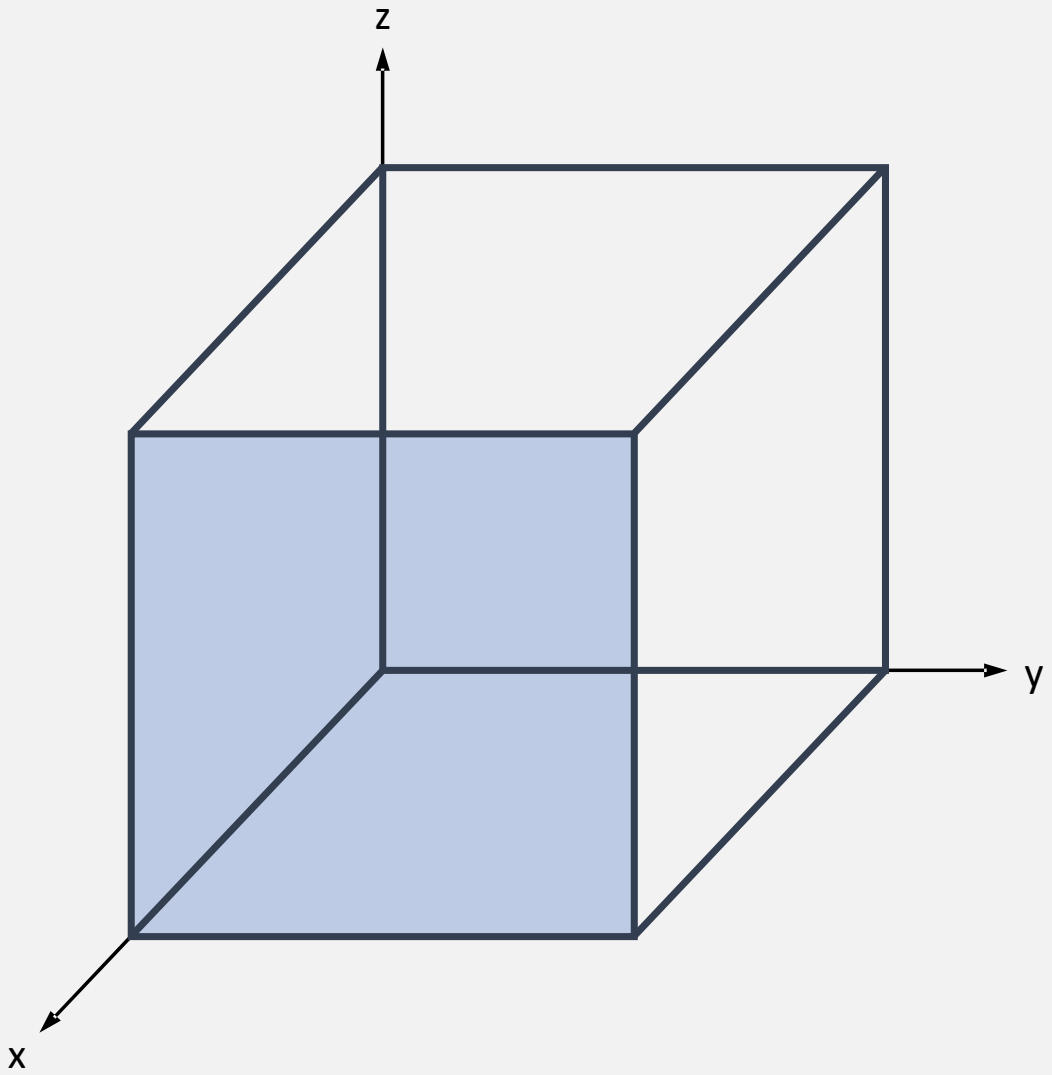
Sets of *(hkl)* planes are simplified to the lowest common factor*:*

$\{2\bar{2}0\} \Rightarrow \{1\bar{1}0\}$



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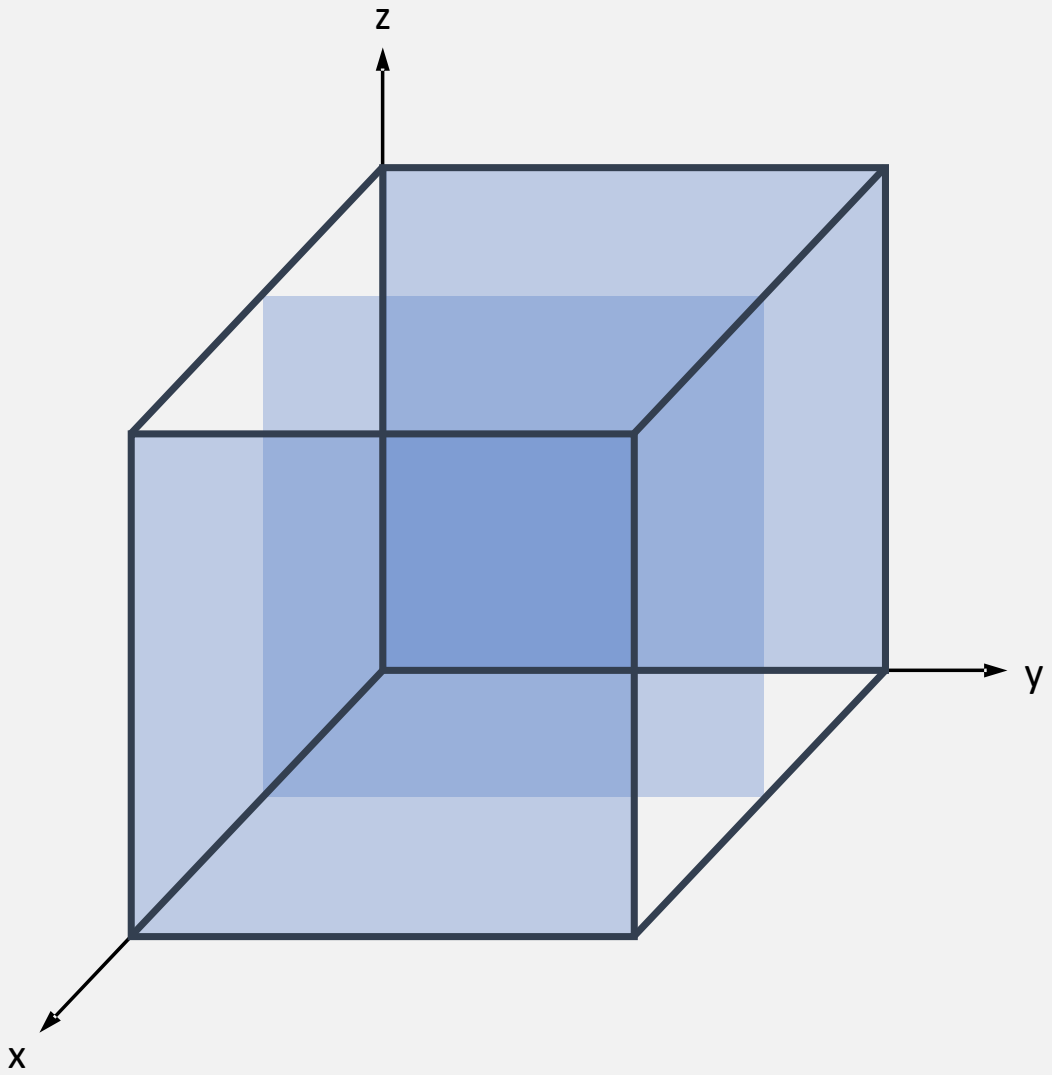
Miller indices in 3D



(100)

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Miller indices in 3D

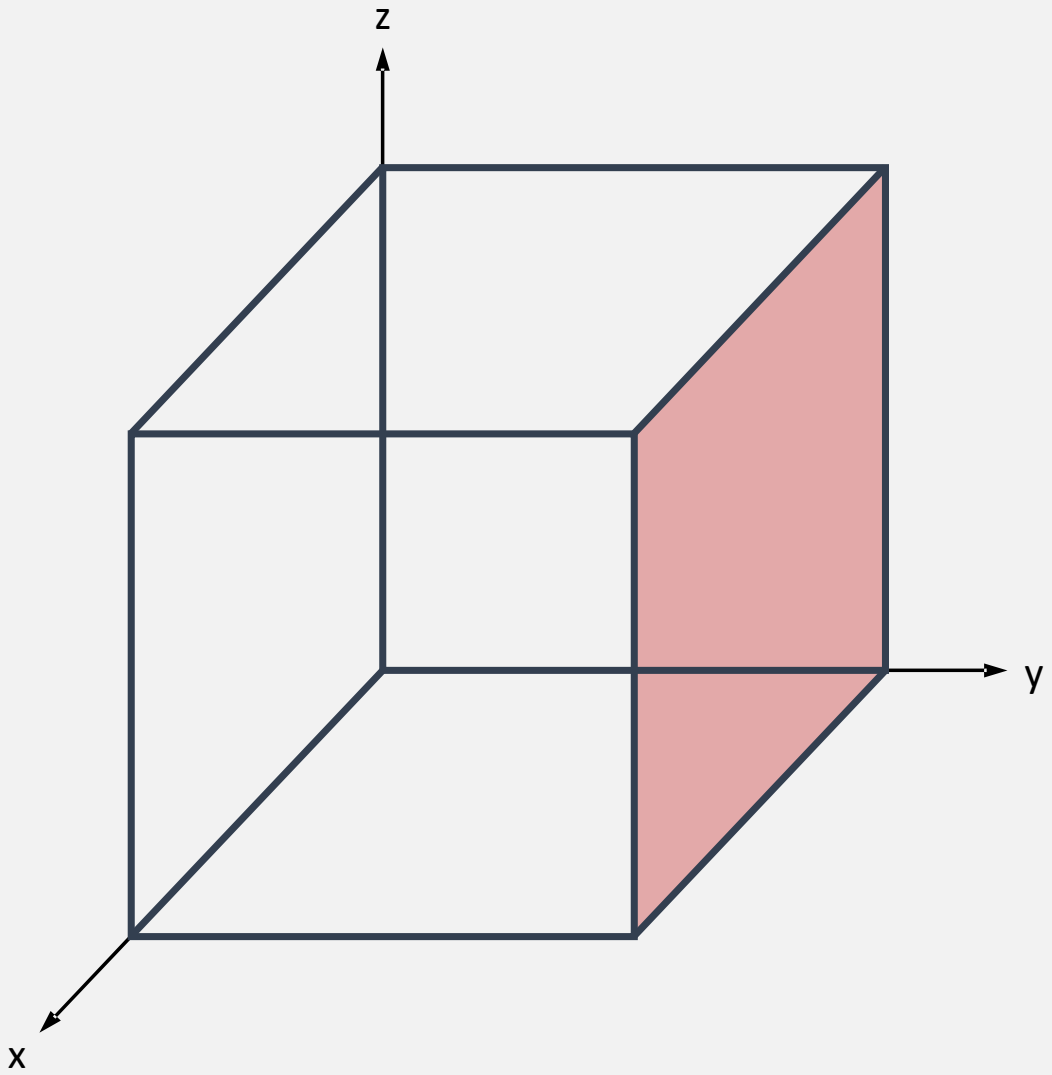


(200)

{100} set of planes

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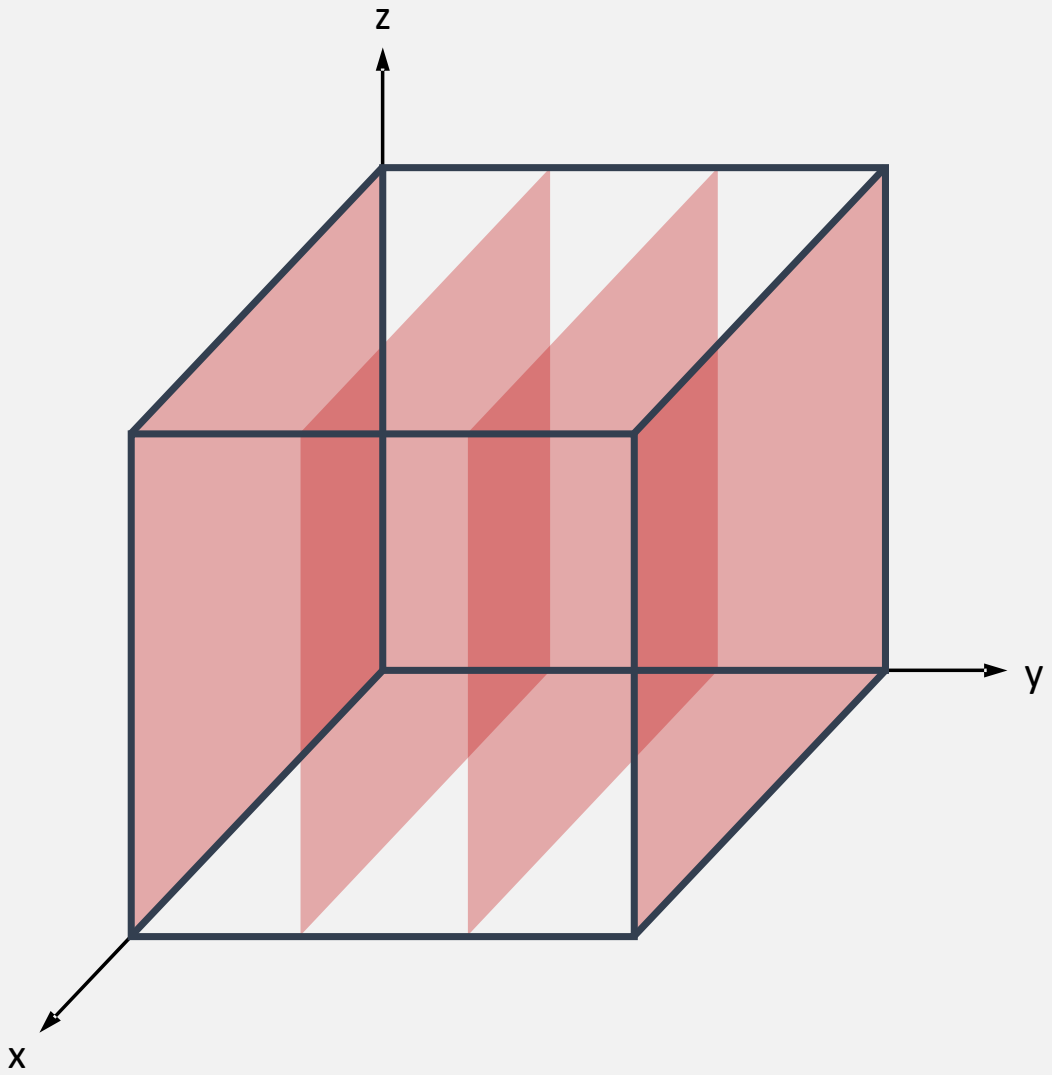
Miller indices in 3D



(010)

margaret.hartley@manchester.ac.uk

Miller indices in 3D

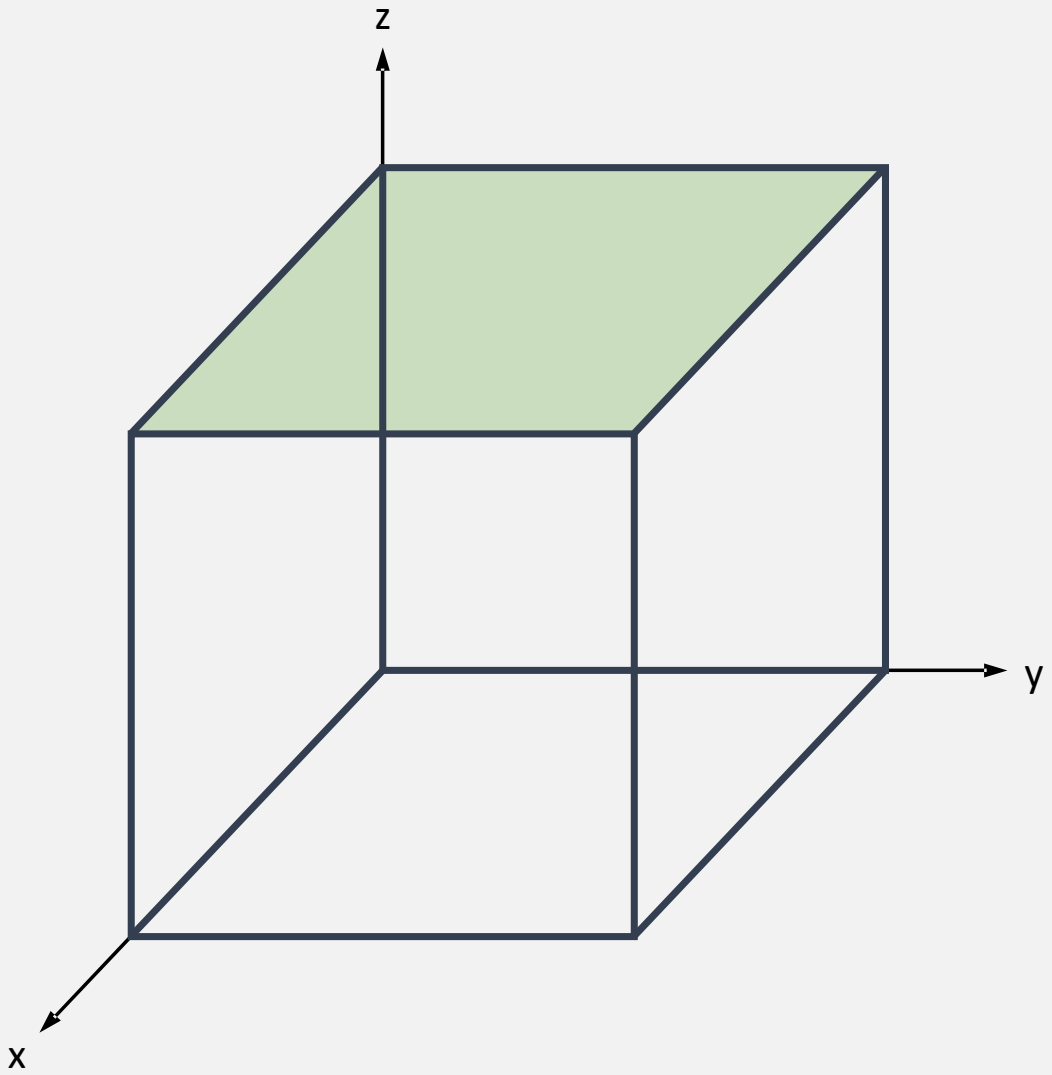


(030)

{010} set of planes

margaret.hartley@manchester.ac.uk

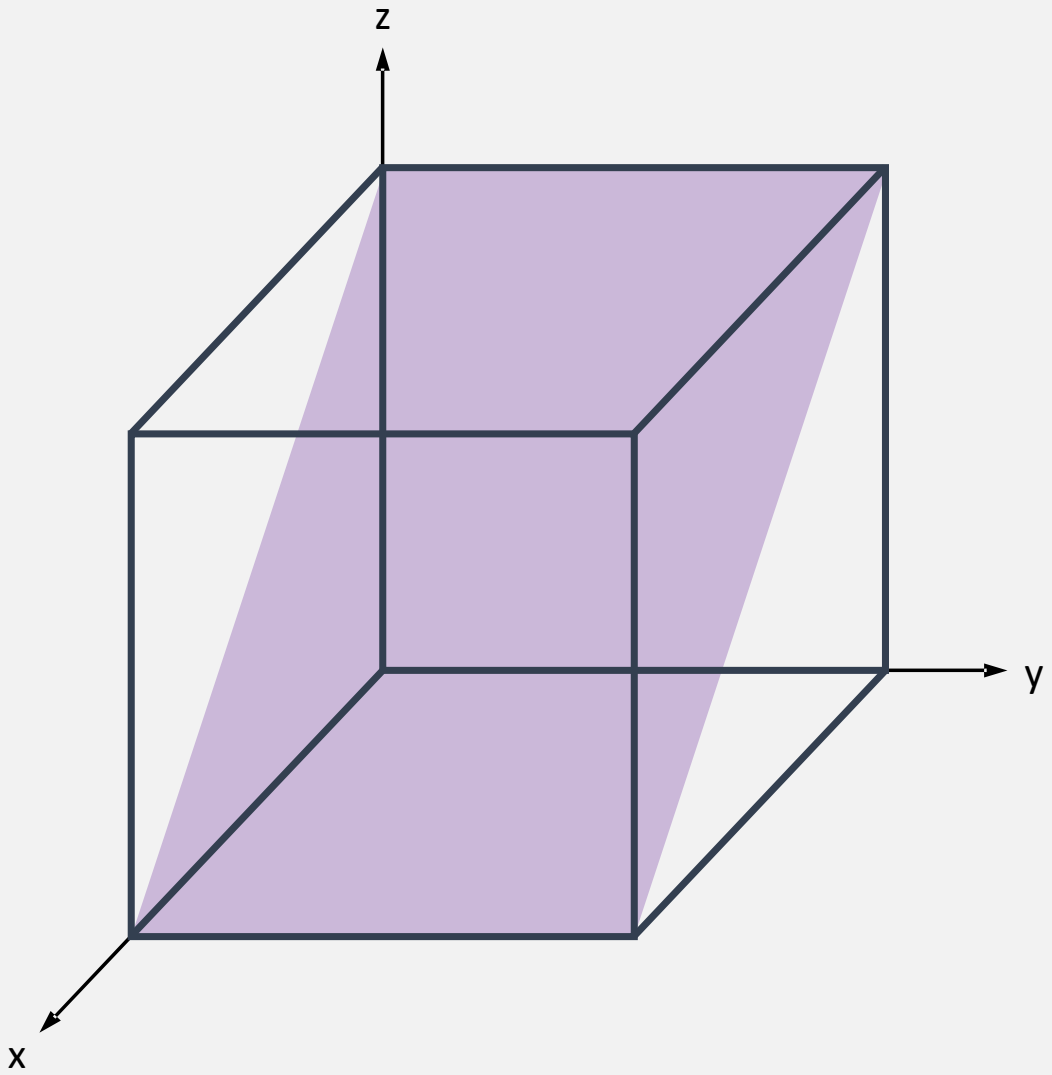
Miller indices in 3D



(001)

margaret.hartley@manchester.ac.uk

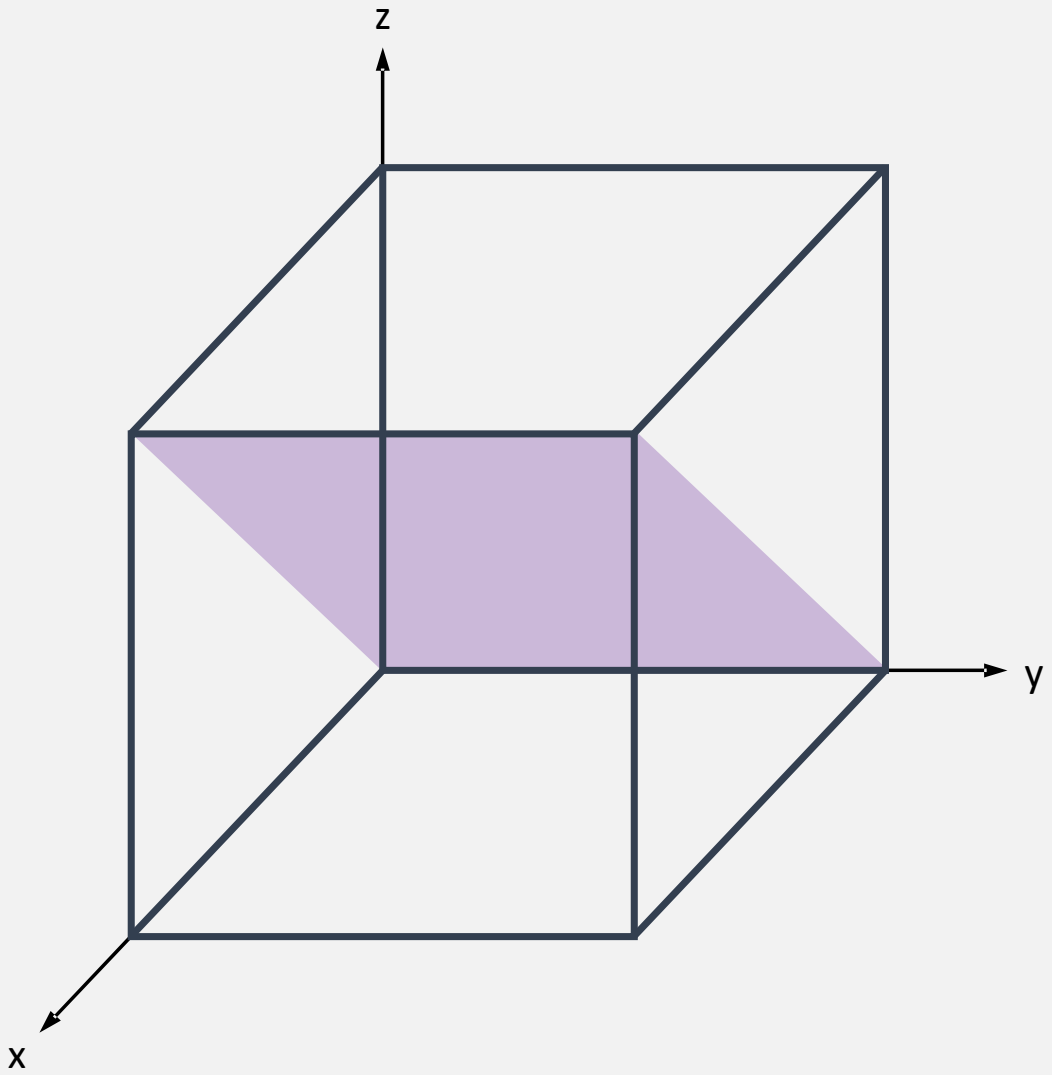
Miller indices in 3D



(101)

margaret.hartley@manchester.ac.uk

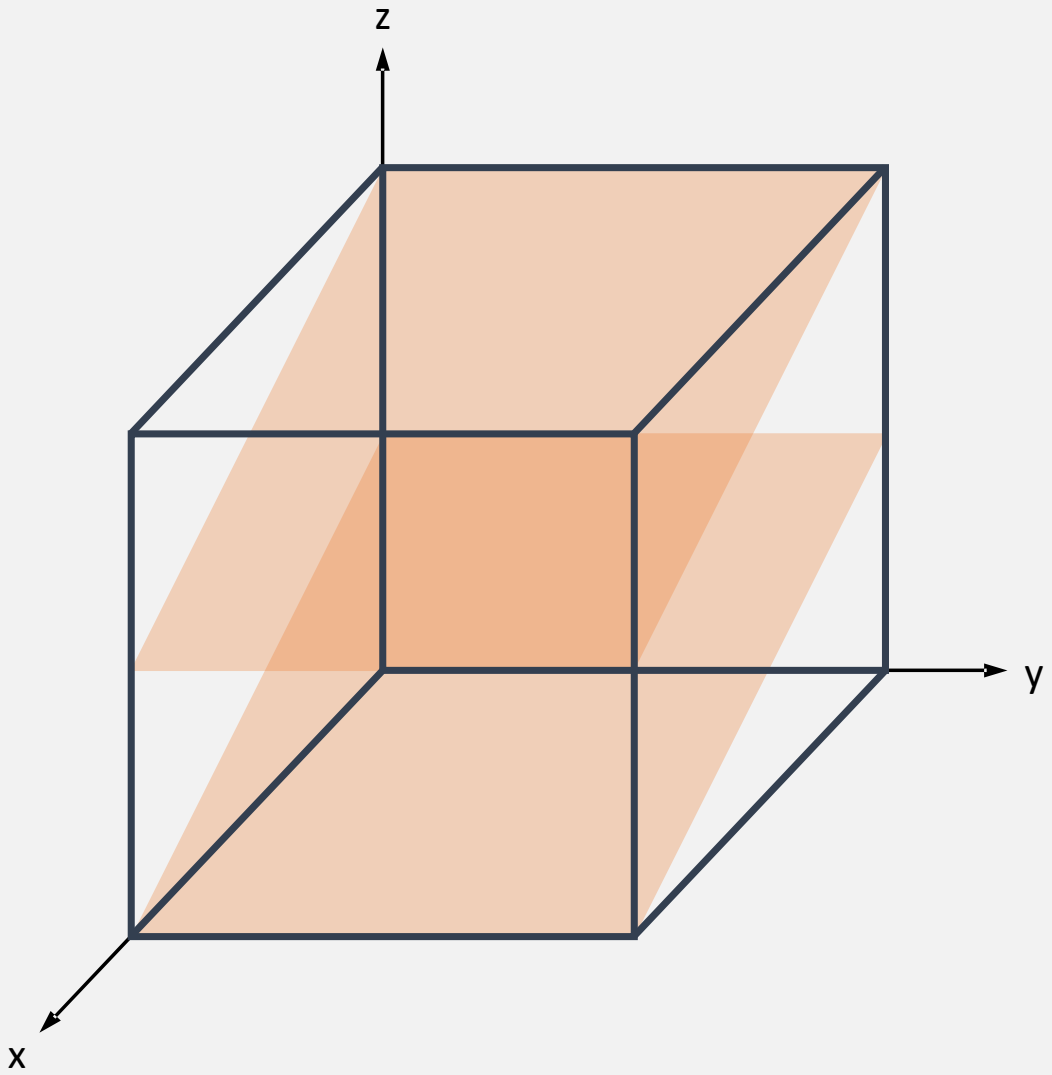
Miller indices in 3D



$(10\bar{1})$

margaret.hartley@manchester.ac.uk

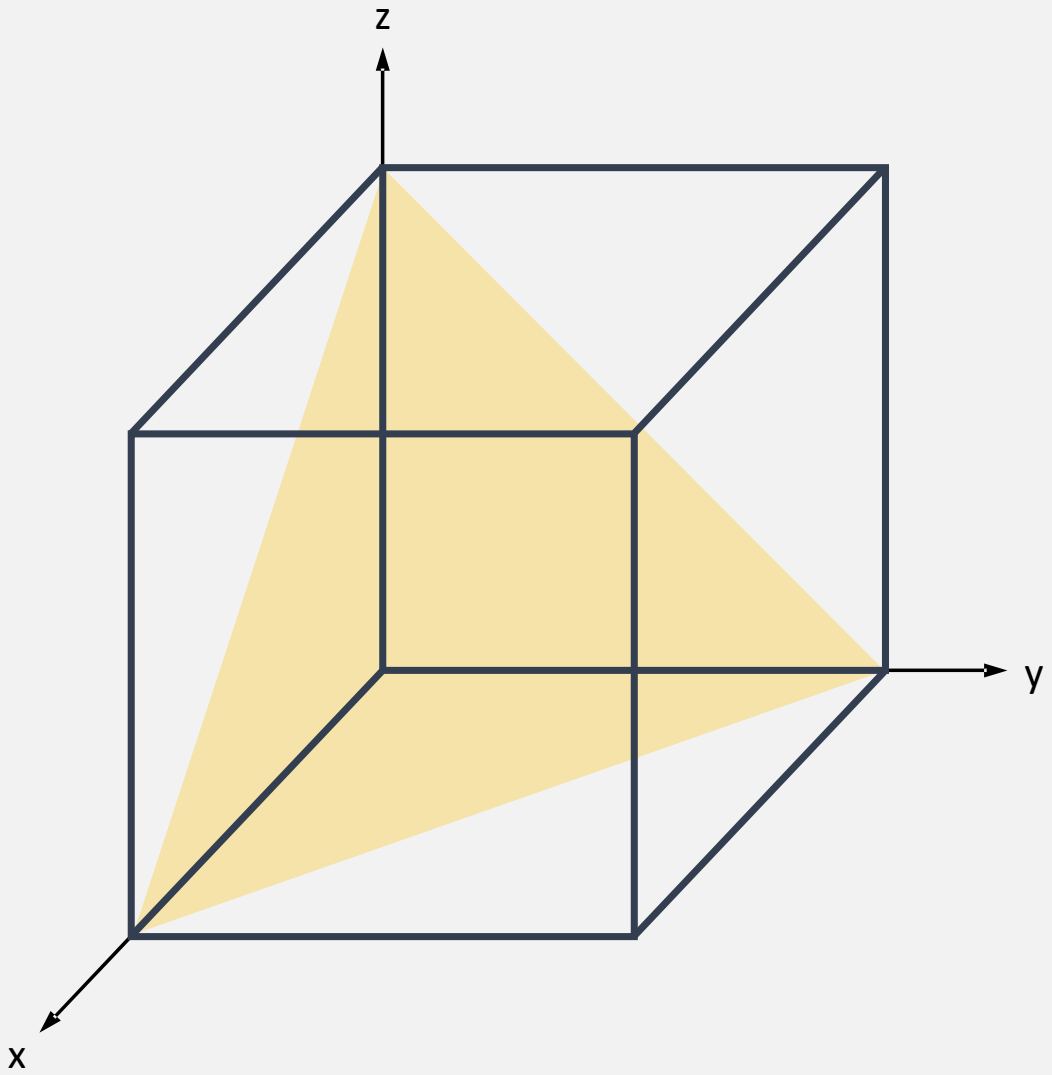
Miller indices in 3D



{102} set of planes

margaret.hartley@manchester.ac.uk

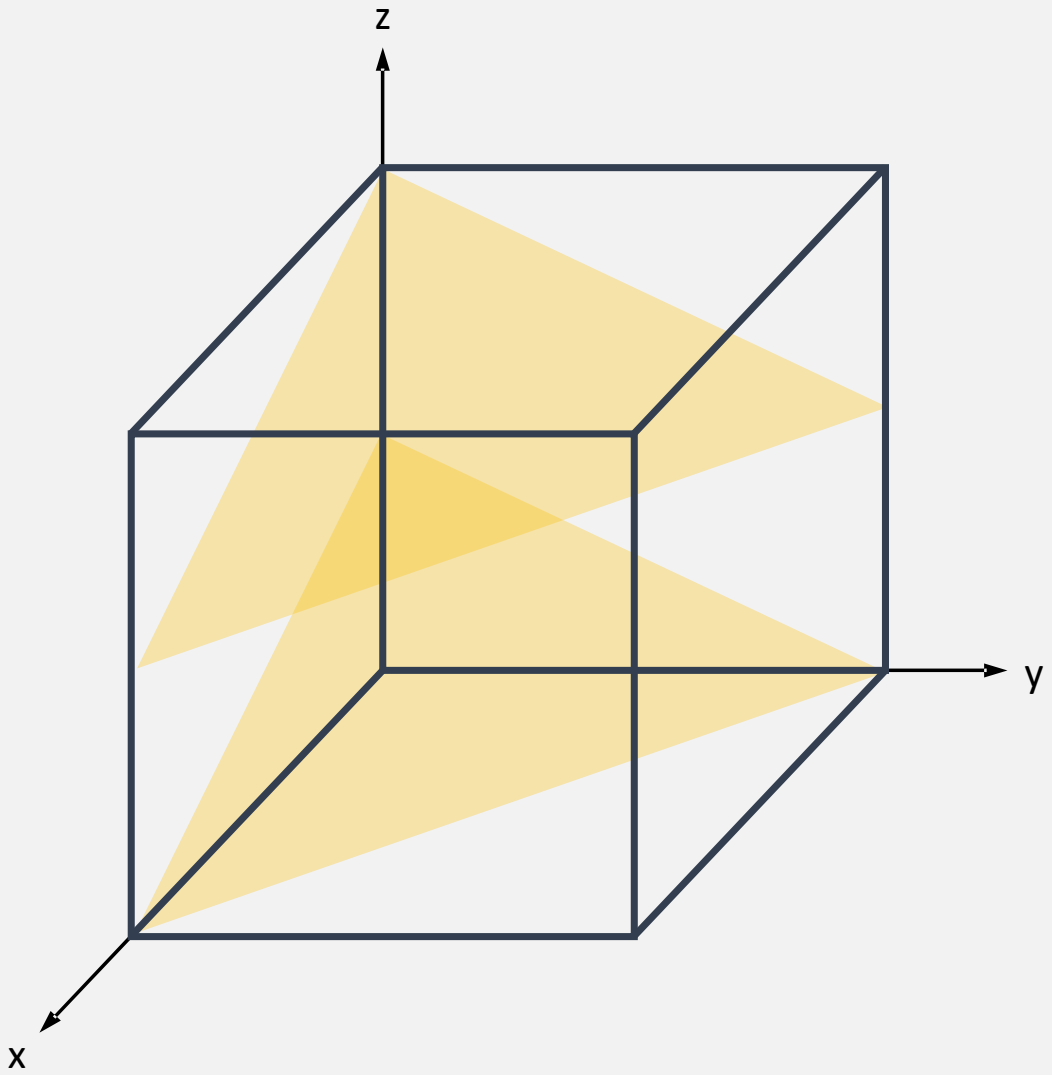
Miller indices in 3D



(111)

margaret.hartley@manchester.ac.uk

Miller indices in 3D



{112} set of planes

margaret.hartley@manchester.ac.uk

Symmetry and the seven crystal systems

4.1 Miller indices

4.2 Cleavage and twinning in minerals



margaret.hartley@manchester.ac.uk

Planar features in minerals

cleavage



halite



biotite

twinning



augite



pyrite

twinned plagioclase in thin section



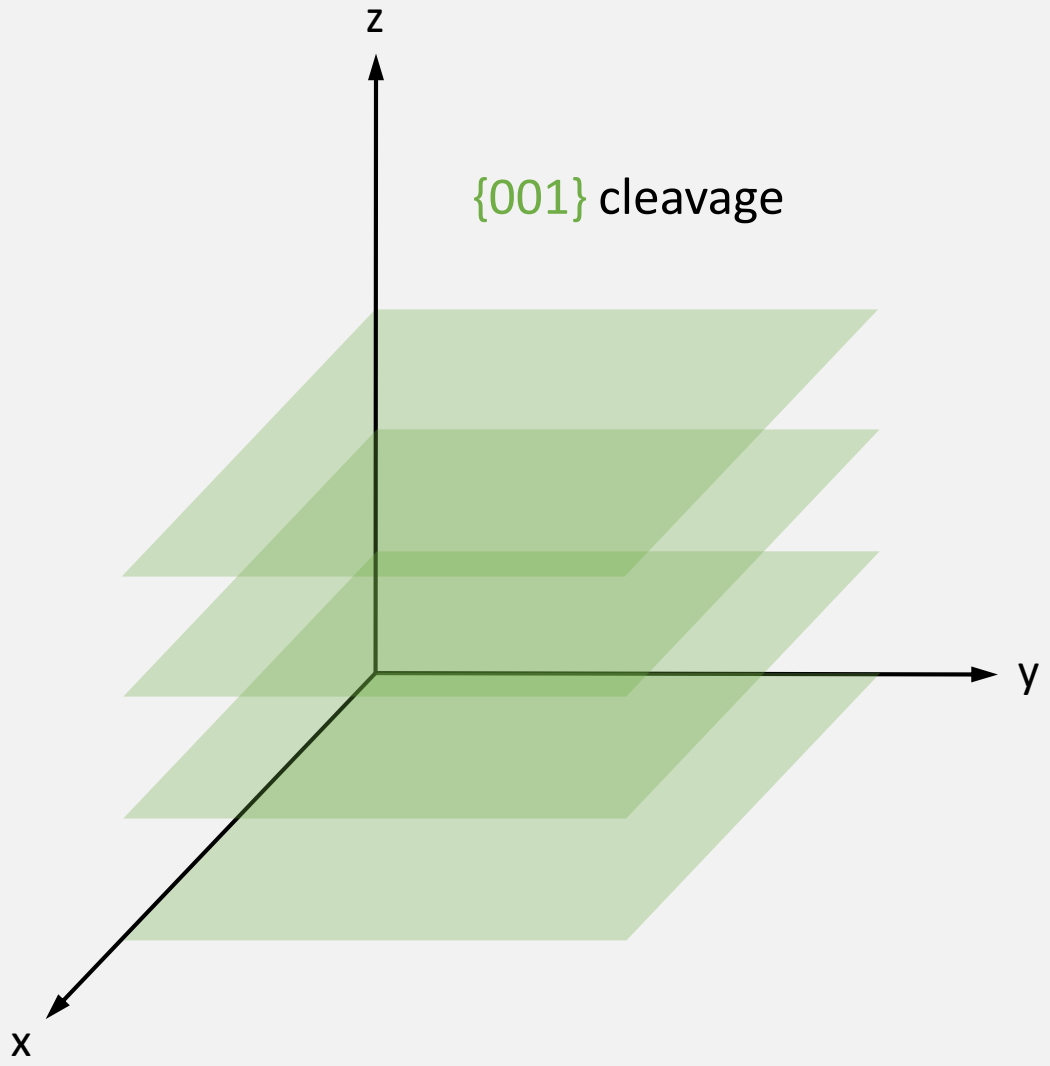
margaret.hartley@manchester.ac.uk

Cleavage



1 perfect basal cleavage

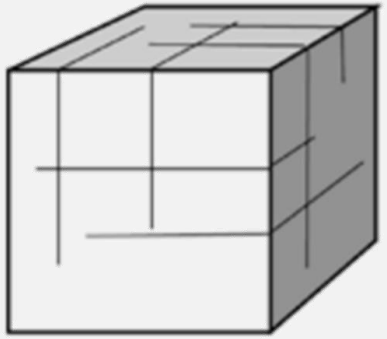
e.g. biotite, muscovite



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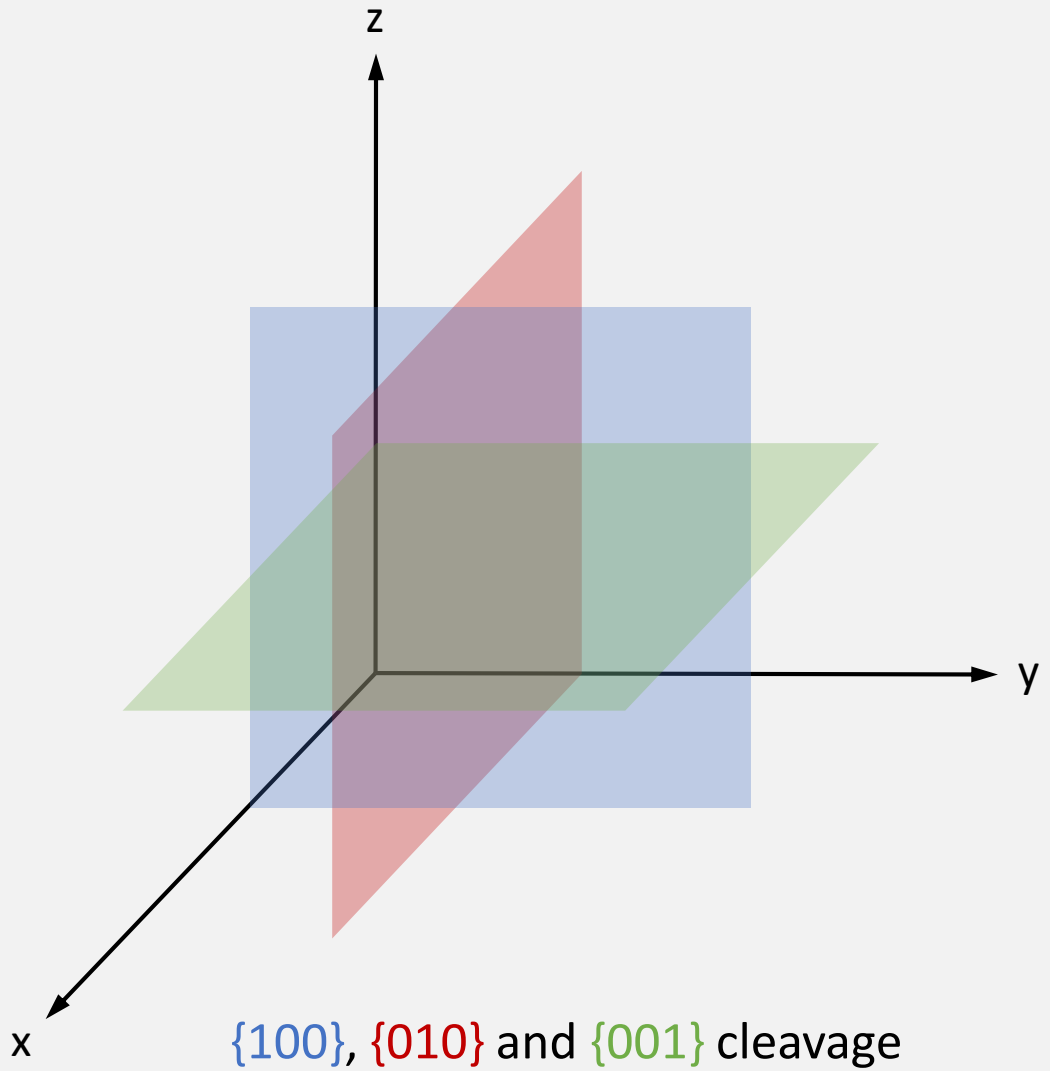
Cleavage





3 cleavages at right angles

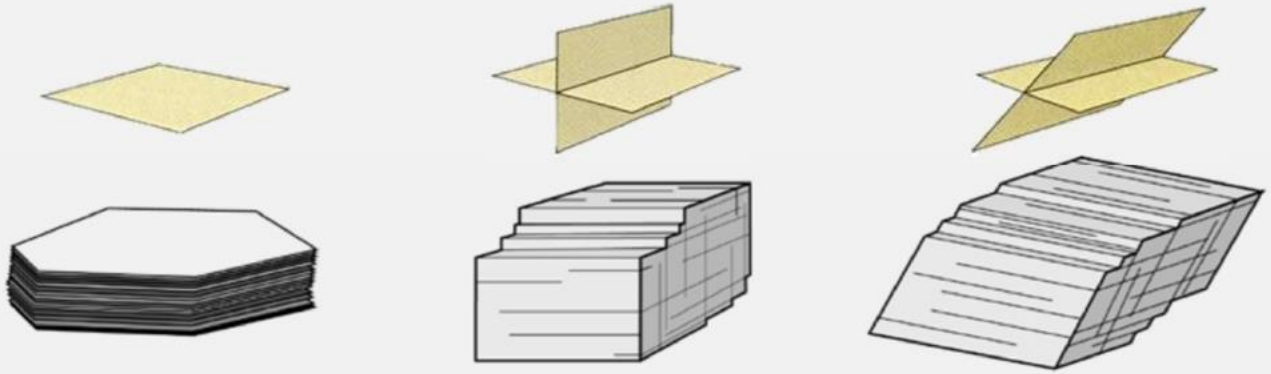
e.g. halite, galena



{100}, {010} and {001} cleavage

margaret.hartley@manchester.ac.uk

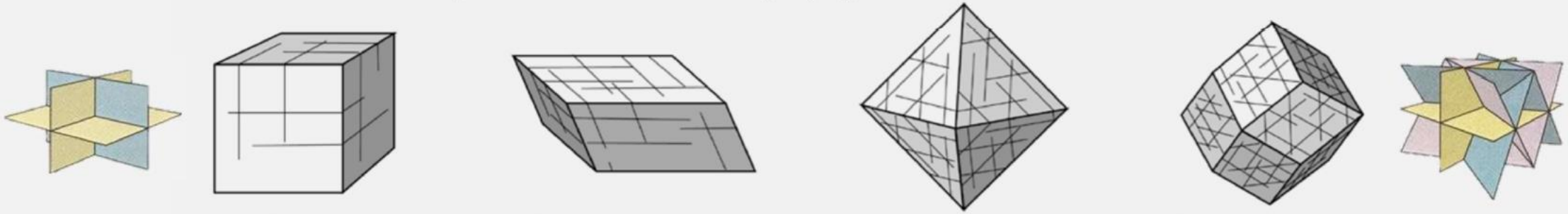
Cleavage



Basal
(1 cleavage)
biotite, muscovite

Prismatic
(2 cleavages at right angles)
feldspars, augite

Non-prismatic
(2 cleavages not at right angles)
hornblende



Cubic
(3 cleavages at right angles)
halite, galena

Rhombohedral
(3 cleavages, none at right angles)
calcite

Octahedral
(4 cleavages -- 8 faces)
fluorite

Dodecahedral
(6 cleavages -- 12 faces)
sphalerite



margaret.hartley@manchester.ac.uk

Symmetry and the seven crystal systems

You should now be able to:

- Use Miller indices to draw oriented planes on a lattice
- Work out the Miller index of planar features from diagrams
- List different kinds of planar features in minerals



margaret.hartley@manchester.ac.uk

Symmetry and the seven crystal systems

5.1 Worked example: twinning in hornblende



margaret.hartley@manchester.ac.uk

Worked example: twinning in hornblende



Twin planes along {100}

Lattice parameters:
a = 10 Å
b = 18 Å
c = 5 Å
β = 105°

"Draw a projection looking down the y axis, showing the unit cell dimensions and the orientation of the twin planes."

margaret.hartley@manchester.ac.uk

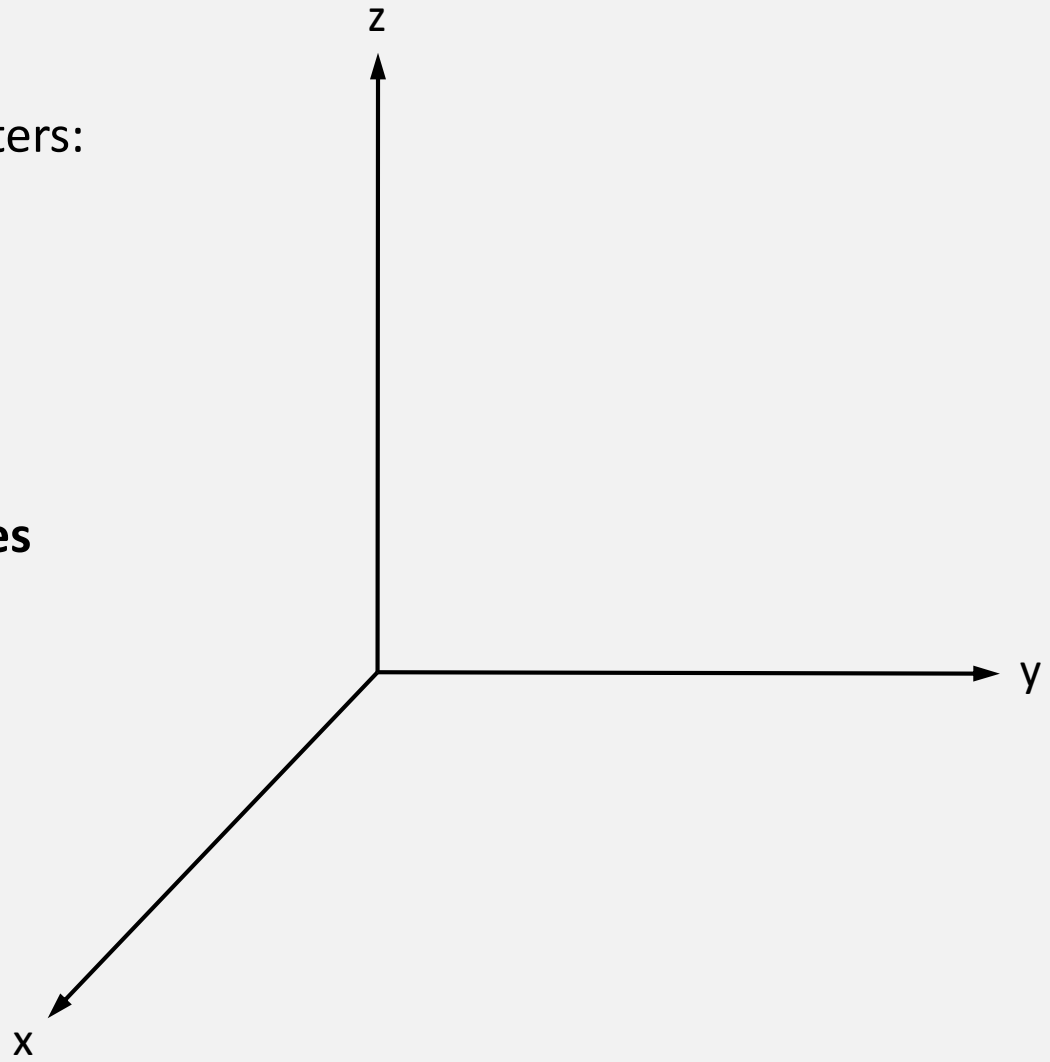
Worked example: twinning in hornblende



Twin planes along {100}

Lattice parameters:
a = 10 Å
b = 18 Å
c = 5 Å
β = 105°

1. Draw the axes



margaret.hartley@manchester.ac.uk

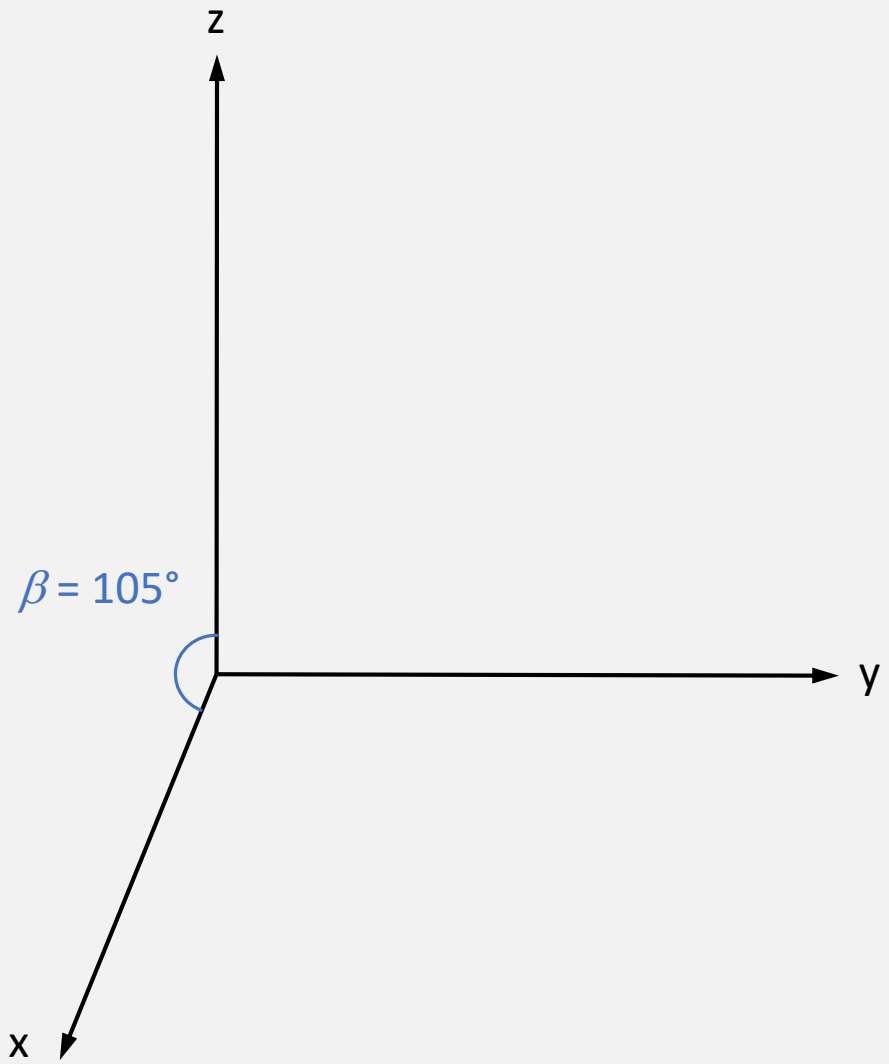
Worked example: twinning in hornblende



Twin planes along {100}

Lattice parameters:
a = 10 Å
b = 18 Å
c = 5 Å
β = 105°

1. Draw the axes



margaret.hartley@manchester.ac.uk

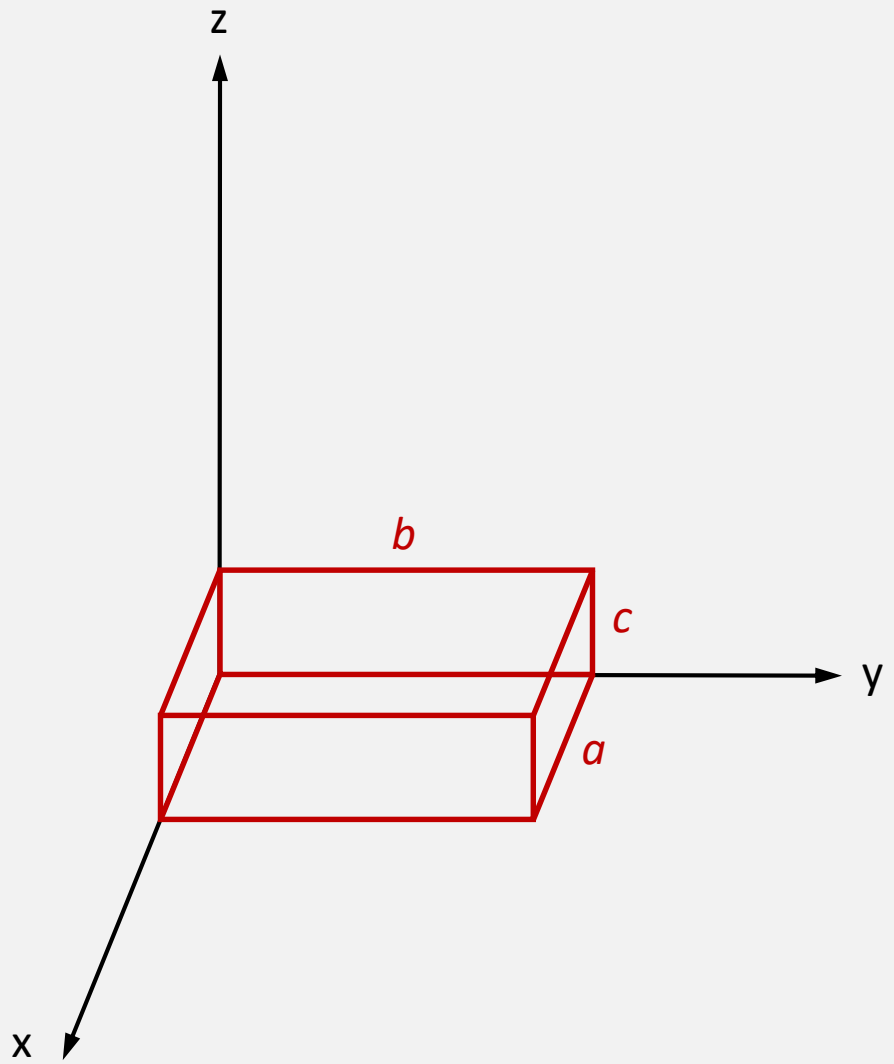
Worked example: twinning in hornblende



Twin planes along {100}

Lattice parameters:
a = 10 Å
b = 18 Å
c = 5 Å
β = 105°

1. Draw the axes
2. Draw the unit cell



margaret.hartley@manchester.ac.uk

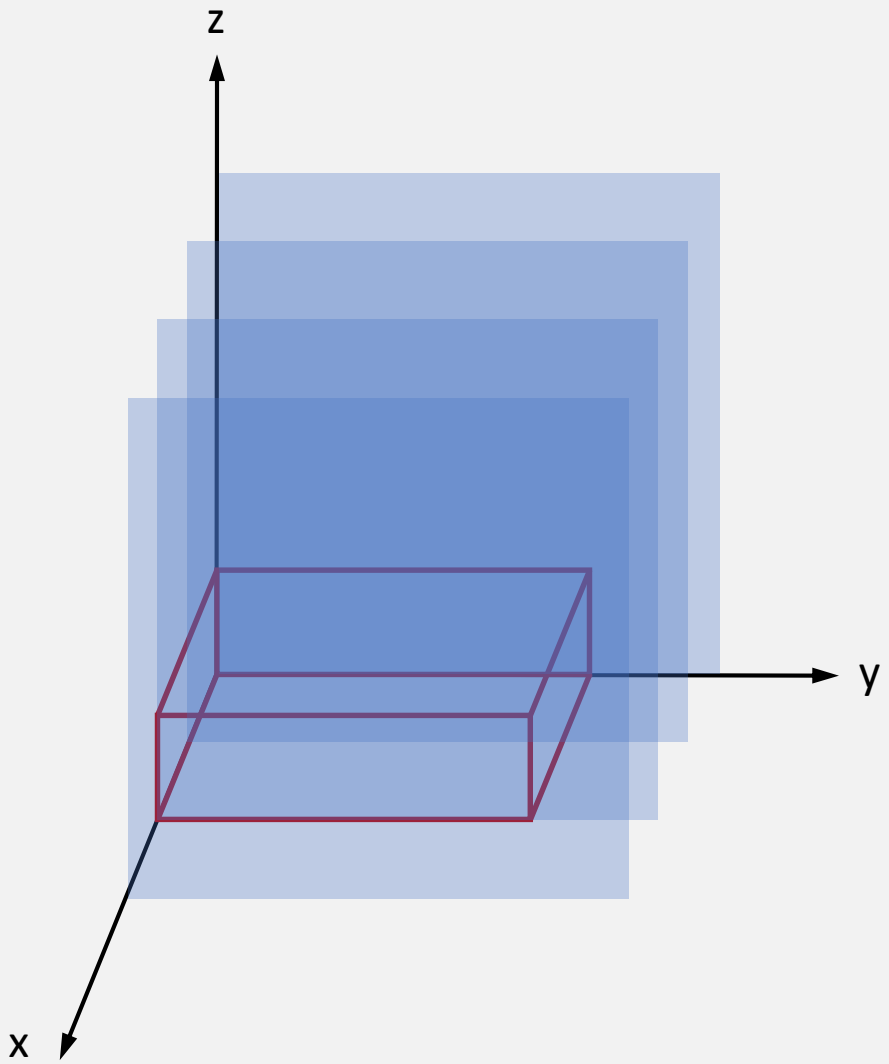
Worked example: twinning in hornblende



Twin planes along {100}

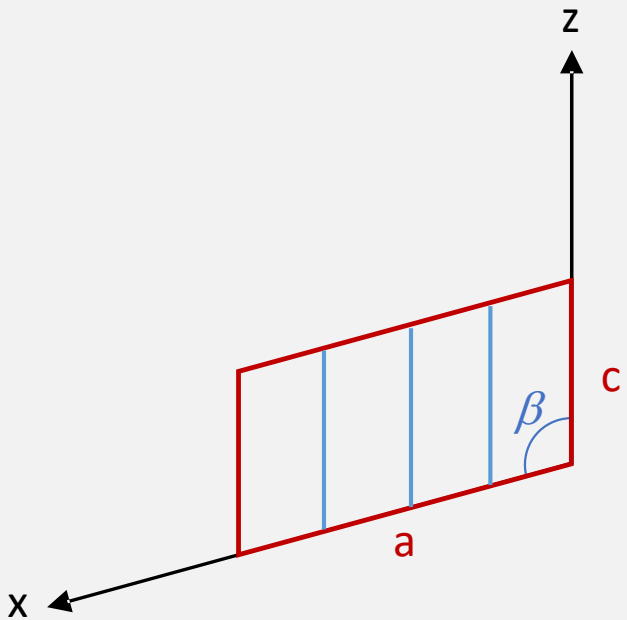
Lattice parameters:
a = 10 Å
b = 18 Å
c = 5 Å
β = 105°

1. Draw the axes
2. Draw the unit cell
3. Add {100} twins



margaret.hartley@manchester.ac.uk

Worked example: twinning in hornblende



Lattice parameters:

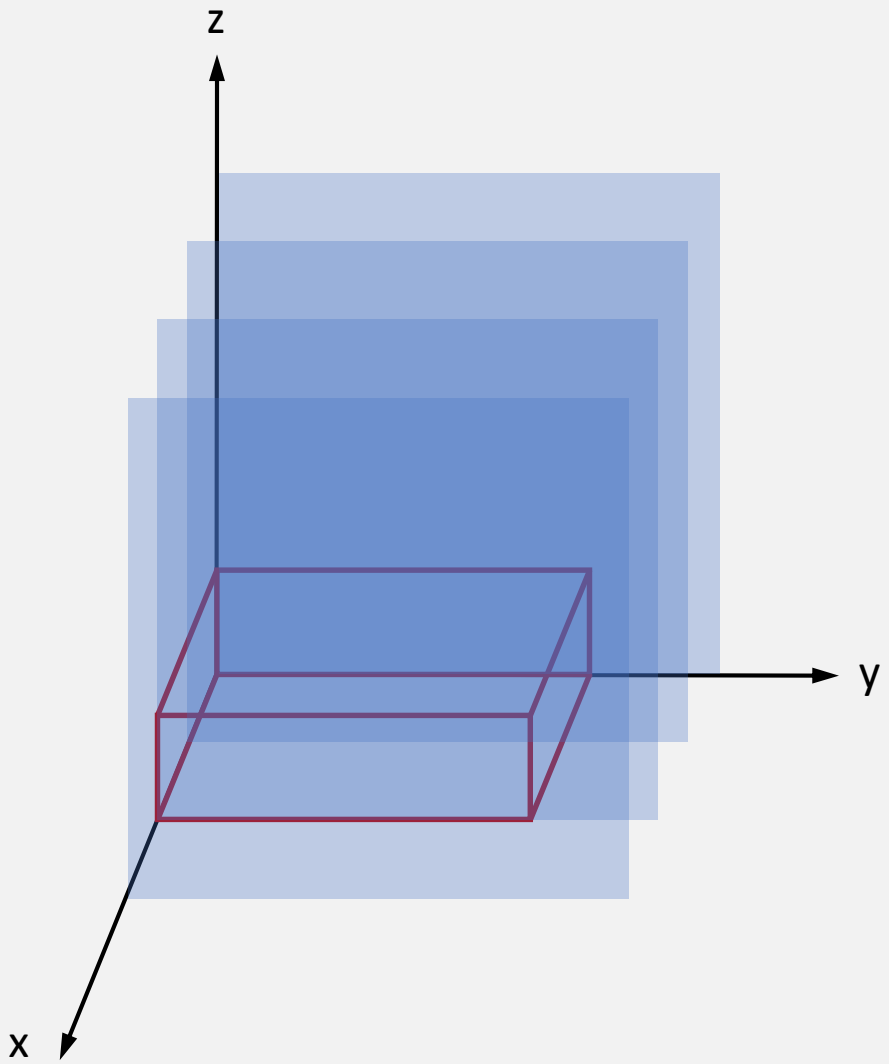
a = 10 Å

b = 18 Å

c = 5 Å

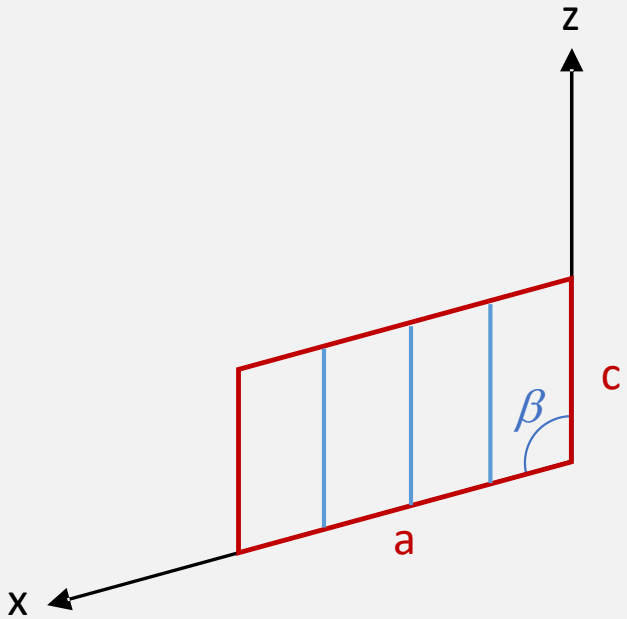
β = 105°

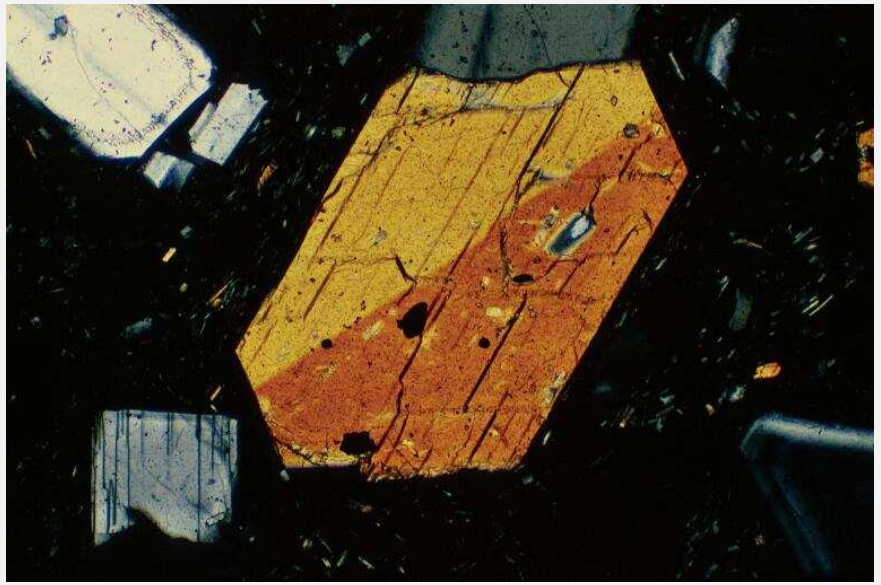
1. Draw the axes
2. Draw the unit cell
3. Add {100} twins
4. **Draw projection ⊥ y**



margaret.hartley@manchester.ac.uk

Worked example: twinning in hornblende





margaret.hartley@manchester.ac.uk

Symmetry and the seven crystal systems

You should now be able to:

- Draw 2D and 3D diagrams of a crystal lattice using information about the lattice parameters
- Use Miller indices to draw planar features on 2D and 3D diagrams of a unit cell



margaret.hartley@manchester.ac.uk